

Transmission of Tariff Shocks via Institutional Investors^{*}

Yiqun Li[†]

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[†]Boston University, Questrom School of Business, Email: yl5125@bu.edu

Abstract

This paper examines how U.S. institutional investors amplify the effects of U.S. tariff shocks on U.S. and foreign firms. Using firm-level data from 2016–2023, I classify firms into directly targeted foreign exporters (*Exposed*), firms indirectly linked to exposed firms via customer–supplier relationships (*Customer* and *Supplier*), and product-market competitors of exposed firms (*Competitor*). I find that tariff shocks are largely unanticipated: pre-shock cumulative abnormal returns (CARs) are statistically indistinguishable from zero. Following tariff shocks, CARs decline for U.S. customer and supplier firms but rise for U.S. competitors. Beyond asset prices, tariff shocks generate economically meaningful effects in firm outcomes. For foreign exposed firms, Tobin’s Q falls by 0.2 in the shock quarter and remains depressed thereafter. Moreover, I identify a novel financial amplification channel operating through U.S. institutional trading, and find U.S. institutions sell off exposed, customer, and supplier firms. A one-standard-deviation decline in U.S. institutional holdings of U.S. customer firms during tariff-shock quarters leads to an additional 2.2 percentage points reduction in sales growth and a 2.4 percentage points contraction in R&D growth. By contrast, increased U.S. institutional holdings lead to higher profitability and stronger capital expenditure growth among U.S. competitors in the long run. Finally, I show that tariff shocks also reshape firms’ real operations: affected U.S. firms experience weaker foreign sales growth and shift toward more domestic supply-chain partners. These findings imply that institutional portfolio reallocation both amplifies and prolongs the real effects of tariff shocks, while tariff shocks also induce firms to reshape their supply-chain structure and reallocate sales activity toward domestic markets.

Keywords: U.S. tariff shocks; institutional investors; supply chain; spillover effect; asset prices; firm outcomes

1 Introduction

Recent U.S. tariff actions have become a major source of concern for firms and investors, both domestically and internationally, heightening global economic tensions. Uncertainty about how these tariffs will reshape global supply chains can have adverse effects on firm performance. A substantial body of international trade examines how trade tariffs influence firms' investment, hiring, and profitability. In this paper, I investigate how institutional investors, such as mutual funds and hedge funds, amplify the effects of tariff shocks on U.S. and foreign firms, and I document the critical role these investors play in transmitting tariff shocks throughout the economy.

Prior studies highlight financial intermediaries and investors as central channels for risk sharing and shock transmission (Khwaja and Mian, 2008; Kruttli et al., 2022; Hale et al., 2020; Giannetti and Saidi, 2019). When tariffs are implemented, however, an important question arises: do U.S. institutional investors amplify or mitigate the effects of U.S. tariff shocks on asset prices and firms' outcomes? U.S. investors may respond in two distinct ways. They may shift capital away from tariff-exposed foreign firms, sending adverse information signals that depress equity prices and weaken fundamentals. Alternatively, as tariff shocks drive down prices, they may increase their holdings of undervalued yet fundamentally solid firms, helping stabilize valuations and dampen the real impact of the shocks.

Tariff shocks provide an ideal laboratory for examining how U.S. institutional investors respond to economic and geopolitical disruptions. As shown in Figure 1, a growing share of U.S. imports from China became subject to tariffs during the trade-war period, creating large and heterogeneous exposure across firms. Moreover, Figure 2 shows that U.S. firms' domestic sales increase sharply during the shaded period, when many tariff announcements were made, suggesting a broader reorientation of firm activity toward the domestic market. As a key policy instrument in trade disputes, tariffs operate as taxes on imported goods: they raise input and product costs, distort relative prices, weaken the comparative advantage of targeted foreign firms, and propagate shocks through global supply chains. Unlike

aggregate measures of economic or geopolitical uncertainty (Baker et al., 2016; Caldara and Iacoviello, 2022), tariff shocks generate plausibly exogenous, product-level variation that can be mapped to firm-level exposure, allowing for cleaner identification of their real effects (Fajgelbaum et al., 2020). Consistent with this identification, equity markets respond sharply and asymmetrically to tariff events: U.S. customers and suppliers of exposed foreign firms earn negative CARs, whereas U.S. competitors earn positive CARs over event windows. This asymmetric market response motivates my focus on institutional portfolio reallocation as a financial transmission channel, and provides a setting to quantify whether U.S. institutional investors’ trading decisions extend the effects of tariff shocks beyond short-run valuation responses to firms’ longer-run real outcomes.

Against this backdrop, the role of U.S. investors—especially U.S. institutional investors—has become increasingly important in shaping global financial dynamics. This growing role is reflected in Figure 3, which documents a robust pattern in the composition of U.S. cross-border portfolio positions, using data from the IMF Portfolio Investment Positions by Counterpart Economy dataset (formerly Coordinated Portfolio Investment Survey, or CPIS). For the United States, equity assets held abroad exhibit a clear upward trend with episodic fluctuations, whereas non-equity holdings (mainly debt) remain comparatively stable and lower. Because the vertical axis is normalized by U.S. GDP, a rising equity-to-GDP ratio implies that the absolute level of U.S. foreign equity holdings grows very rapidly. Hence, even without further financial deepening, continued economic expansion naturally amplifies U.S. exposure to global equity markets, strengthening its role as a global capital supplier. If, moreover, the equity share itself continues to rise, the growth in foreign equity positions will outpace GDP growth, further reinforcing the U.S. footprint in global financial intermediation.

Figure 4 extends this perspective by documenting that in most major economies—especially France, India, and Japan—the share of domestic equity held by U.S. investors has exhibited a significant upward trend. This pervasive pattern is consistent with a broader global shift toward equity-based cross-border ownership and indicates a deepening of international

financial integration. The growing equity participation of U.S. investors, largely institutional, tightens cross-market linkages and heightens the sensitivity of local markets to global capital flows and shocks. Taken together, Figures 3 and 4 suggest a structural increase in cross-border equity interdependence and the expanding role of institutional investors in transmitting and amplifying global financial shocks.

Moreover, tariff shocks have meaningful implications not only for directly targeted firms but also for their broader supply-chain networks. Directly affected firms often experience declines in sales and profitability, leading to cuts in investment and R&D. Spillover firms—those linked through supply chains or investor networks—may also suffer deteriorating fundamentals as institutional investors, particularly U.S. investors, adjust their portfolios through shorting or capital reallocation. At the same time, tariff shocks can create relative winners. Product-market competitors of exposed firms may benefit from demand reallocation and reduced competitive pressure, improving profitability and stimulating investment. Consistent with this mechanism, institutional trading toward competitor firms can operate in the opposite direction: reallocations of capital into competitors can reinforce these gains by supporting higher net margins and stronger capital expenditures. As illustrated in Figure 5, this framework allows us to examine not only the direct consequences of tariff shocks but also the indirect effects transmitted through supply-chain linkages and the financial intermediation channel through which U.S. investors propagate shocks across firms and markets.

To empirically investigate these mechanisms, I build on the dataset and empirical framework of Fajgelbaum et al. (2020) and Bown and Kolb (2022), extending the scope of the trade war beyond its initial phase in 2018–2019 to incorporate all subsequent tariff events. In addition, I use the Temporary Trade Barriers Database (TTBD), which contains detailed information on the U.S. anti-dumping (AD) and countervailing duty (CVD) measures, including the precise set of affected products and the exact timing of duty announcement. These data sources enable me to construct a comprehensive product-level treatment at the HS10 (or HS8) level, identifying both the directly targeted goods and the specific dates

when tariffs were announced and imposed. By merging this information with Panjiva customs shipment records, I am able to trace which foreign firms exported the affected products to the United States, thereby generating a granular, firm-level measure of exposure to the U.S. tariff shocks.

I then link these exposure measures to FactSet 13F ownership data, which report U.S. institutional investors' holdings in global firms. This novel integration of trade and financial data allows me to trace how U.S. institutions rebalance their equity portfolios in response to tariff shocks, and to quantify the real consequences of those reallocations. By observing position changes around tariff announcements, I document that U.S. institutions systematically reduce their holdings in directly exposed foreign exporters and in foreign spillover firms connected to exposed exporters through supply chains. In contrast, institutional trading toward competitor firms is more heterogeneous, consistent with some investors reallocating capital toward potential beneficiaries of tariff-induced demand reallocation.

To capture indirect exposure, I incorporate FactSet Revere supply-chain data, which map firm-level customer-supplier linkages. These data allow me to identify firms that are not directly targeted by tariff shocks but are nevertheless exposed through their relationships with affected foreign firms; in other words, I identify both the upstream suppliers (*Supplier*) and downstream customers (*Customer*) of tariff-exposed firms (*Competitor*). I further use Revere competitor links to identify product-market rivals of exposed firms. This extension enables me to distinguish direct tariff incidence from propagation through supply-chain networks and competitive reallocation. Empirically, I show that institutional sell-offs during shock periods amplify subsequent deterioration in customer or supplier firms' performance, consistent with a financial intermediation channel that deepens and prolongs tariff-induced disruptions. At the same time, increases in institutional holdings of competitor firms lead to improvements in profitability and investment, suggesting that capital reallocation can reinforce gains among relative winners. The analysis provides new evidence that U.S. institutional investors are a key mechanism through which tariff shocks propagate across firms

and markets, shaping both the losses among exposed and connected firms and the gains among competitors.

Asset-price evidence indicates that U.S. tariff shocks are largely unanticipated and have sharply asymmetric valuation effects across U.S. customer, supplier and competitor firms. Prior to tariff shocks, cumulative abnormal returns (CARs) for all affected U.S. customer, supplier and competitor firms are statistically indistinguishable from zero. After announcements, CARs diverge: both affected U.S. customer and supplier firms experience persistently negative abnormal returns, whereas U.S. competitor firms exhibit steadily increasing and persistently positive abnormal returns, consistent with negative propagation through supply chains and demand reallocation toward rivals.

Beyond asset prices, tariff shocks generate economically meaningful and heterogeneous real effects, and U.S. institutional investors play a central amplifying role. Directly exposed foreign firms suffer the most severe declines in Tobin's Q and innovation, while spillover effects are smaller but persistent and often lagged; for U.S. firms, the real impacts are concentrated in investment and profitability. Institutional portfolio rebalancing is a key mechanism: U.S. institutions reduce holdings in exposed and spillover firms around shocks, and interaction estimates imply that these sell-offs amplify subsequent contractions in sales growth, R&D, and capital expenditure, whereas increases in holdings reinforce gains among competitor firms. Overall, the institutional amplification channel is quantitatively stronger for U.S. connected firms than for comparable foreign connected firms.

Additional evidence suggests that tariff shocks affect not only firm performance, but also the organization of firms' operations. Following tariff shocks, U.S. firms experience weaker foreign sales growth, indicating a contraction in foreign-market activity through both supplier and customer linkages. At the same time, domesticity mitigates these losses: firms connected to more domestically oriented suppliers or customers suffer smaller declines in sales growth, profitability, and Tobin's Q during the trade-war period. Finally, I find direct evidence of supply-chain reorganization, as U.S. firms increasingly shift toward U.S. partners or partners

with more U.S.-sales-dominant operations. These findings suggest that tariff shocks induce firms to partially restructure their supply chains toward domestic markets, providing a real-side adjustment margin through which firms respond to trade-policy disruptions.

This paper contributes to the emerging literature on geoeconomics and the real effects of trade policy shocks. Prior studies have shown that tariffs depress the performance of directly targeted exporters (Chor and Li, 2024; Fajgelbaum et al., 2020) and transmit indirectly through global supply chains (Huang et al., 2023; Handley et al., 2024) while recent work highlights the role of banks and sanctions in shaping firm responses (Alfaro et al., 2025; Efung et al., 2023; Camous and Cooper, 2021). Yet, researchers still know little about how the U.S. institutional investors themselves transmit and amplify these shocks. By constructing a novel dataset that combines tariff information, customs records from Panjiva, 13F ownership data from FactSet, and firm-level supply-chain linkages from FactSet Revere, I provide the first systematic evidence that U.S. institutional investors not only withdraw capital from directly exposed foreign firms but also actively short the equities of spillover firms when tariff shocks take place. This novel financial amplification channel shows that institutions play a pivotal role in propagating tariff shocks across global supply chains, ultimately shaping the real outcomes of affected foreign firms. My findings therefore bridge the international trade and finance literature, highlighting the importance of financial intermediaries in determining the global reach of protectionist policies. This paper also implies that future research should not only consider supply chains but also incorporate financial chains that transmit shocks through capital reallocation.

The remainder of the paper is organized as follows. Section 2 provides background of each tariffs shock and reviews the related literature on trade policy shocks, supply-chain spillovers, and financial intermediaries. Section 3 describes my data sources. Section 4 outlines the empirical design used to identify the effects of U.S. tariff shocks and U.S. institutional investors on exposed, spillover, and competitor firms, then Section 5 presents my main results on my empirical designs and discusses additional findings. Section 6 further examines whether tariff

shocks lead to supply-chain reconstruction and a shift toward more domestic supply-chain relationships. Section 7 concludes.

2 Background & Literature Review

2.1 Background of Tariff Shocks

The tariffs faced by foreign exporters shipping to the United States are constructed as the sum of the baseline MFN tariff schedule and additional trade policy measures implemented over the 2016–2023 period. In particular, I incorporate anti-dumping (AD) and countervailing duties (CVD), Section 232 tariffs on steel and aluminum, Section 301 tariffs—including those imposed on Chinese imports during the 2018–2019 trade war as well as more recent actions targeting other countries. My construction follows the methodology of Fajgelbaum et al. (2020) and Bown et al. (2022), who compile detailed data on MFN rates and trade-war-related tariff increases. I extend their datasets in two important dimensions. First, we broaden the scope of tariff coverage by systematically integrating AD/CVD cases, which are not fully captured in the original sources. Second, I extend the time horizon to include tariff changes through 2024, drawing on official U.S. government announcements and the World Bank’s Temporary Trade Barriers Database. This approach yields a comprehensive and consistent measure of U.S. tariff shocks at the HS10 (or HS8) product level, forming the foundation for our empirical analysis.

Fajgelbaum et al. (2020) provide a systematic account of the exogenous U.S. tariff shocks during 2018–2019, covering the imposition of Section 232 tariffs on steel and aluminum, as well as multiple rounds of Section 301 tariffs on Chinese imports. Their dataset documents the timing, scope, and product-level coverage (at the HS10 level) of these measures, showing that tariff increases were highly targeted, phased in through successive lists, and fully passed through to U.S. importers. The authors demonstrate that these trade policy actions significantly altered import volumes, raised consumer prices, and created heterogeneous exposure

across sectors.

Bown and Kolb (2022) complement this effort by compiling detailed timelines and product lists for each tariff announcement, tracking both U.S. actions and retaliatory responses. Their dataset captures modifications over time—including exemptions, suspensions, and changes in MFN rates—allowing for a dynamic view of how tariff policy evolved during the conflict. Importantly, their work emphasizes that tariff incidence varied substantially across products and trading partners, providing a granular mapping of which goods and firms were most directly affected. In this paper, I focus exclusively on negative events announced by the U.S. government to ensure cleaner identification.

Building on these two sources, I construct a comprehensive panel of U.S. tariffs at the HS8/HS10 level that extends beyond the 2018–2019 trade war. This allows me to map the exogenous tariff shocks to firms’ export and fundamentals, forming the basis for analyzing how U.S. financial intermediaries respond through portfolio adjustments and how such responses affect the real outcomes of both directly targeted and spillover firms.

Additionally, under WTO rules, member nations may impose AD duties on imports sold at prices below fair market value—i.e., “dumping”—if such practices cause injury to domestic producers. Similarly, CVDs are applied to counteract subsidized imports that materially harm local industry. Both measures are considered types of tariffs and are an integral part of temporary trade barriers.

To systematically capture the application of these measures, I rely on the World Bank’s *Temporary Trade Barriers Database (TTBD)*, which compiles comprehensive case- and product-level information on AD, CVD, and safeguard actions across more than 30 countries from the 1980s through 2024. The database includes key details such as the initiating country, the products affected (at HS-10 or HS-8 levels), the dates of investigation and implementation, and often petitioner identities. Originally launched as the *Global Antidumping Database* in 2005 and later expanded to include CVD data, the TTBD now represents one of the most granular and transparent repository of temporary trade barriers available.

In my analysis, I exploit the TTBD’s detailed timestamps to precisely identify when AD and CVD tariffs targeting specific products take effect. This enables me to merge tariff treatment information with firm-level customs data and financial ownership records. However, I cannot apply safeguard duties from the TTBD due to missing data (Bown, 2014).

2.2 Related Literature

A large and growing literature examines the consequences of tariff shocks for firms, trade flows, and the broader economy. At the firm level, recent studies have documented that tariffs imposed during the U.S.–China trade war created large and heterogeneous effects. Chor and Li (2024) show that Chinese exporters most exposed to U.S. tariffs suffered significant declines in sales, employment, and income. Fajgelbaum et al. (2020) provide systematic evidence that U.S. tariffs led to sharp reductions in imports of targeted goods, full pass-through of tariffs into domestic prices, and aggregate welfare losses. Flaaen et al. (2020) study the U.S. tariffs on washing machines and find that they raised consumer prices not only for washing machines but also for related goods such as dryers, illustrating how trade policy shocks extend beyond directly taxed products. Together, this body of work establishes tariffs as salient, product-level shocks with immediate consequences for firms and consumers.

Another branch of the literature highlights how exporters adjust to tariff shocks by reallocating trade flows across markets. Fajgelbaum et al. (2024) demonstrate that bystander countries such as Vietnam and Mexico expanded their exports as China lost market share in the United States, highlighting trade diversion as a key adjustment margin. Jiang et al. (2023) show that U.S. tariffs sharply reduce Chinese exports to the United States, but overall exports remain stable as firms divert sales toward larger, richer, and geographically closer third-country markets. Benguria and Saffie (2024) show that retaliatory tariffs cut U.S. exports to targeted markets but were largely offset by gradual diversion to similar third-country destinations—especially in high-leverage, low-stickiness industries—leaving total exports little changed overall. These findings suggest that tariffs reshape not only the behavior of

directly targeted firms but also the global allocation of exports across countries.

Beyond direct impacts, tariff shocks propagate through production networks and affect firms not directly targeted by trade policy. Barrot and Sauvagnat (2016) show that shocks to suppliers can significantly reduce downstream firms’ output and market value, with effects magnified by concentration in the supply base. Handley et al. (2024) demonstrate that U.S. tariffs on Chinese inputs disrupted the performance of American buyers, while Huang et al. (2023) find that U.S. firms with greater exposure to Chinese supply chains experienced larger market value losses, which then spilled over to their suppliers and customers. These literature make clear that tariff shocks have consequences well beyond the directly exposed firms, affecting entire production networks.

Additionally, a broader literature situates tariffs and sanctions within the domain of geo-economic policy, which links geopolitical rivalry to economic and financial outcomes. Ahn and Ludema (2020) show that targeted sanctions in 2014–2016 led to large declines in sanctioned Russian firms’ revenues, assets, and employment—especially in service-input-intensive sectors—while state protection of “strategic” firms shifted a substantial share of the burden onto the public sector. Crosignani et al. (2024) use U.S. export controls as plausibly exogenous shocks, the paper shows that exposed domestic mutual funds suffer higher volatility and lower returns; active funds quickly sell affected and China-linked stocks and re-balance toward lottery/size/value exposures to partially offset losses, while passive funds do not adjust and under-perform. Hinz and Monastyrenko (2022) show that Russia’s 2014 food-import embargo raised prices of banned goods and, via input–output spillovers, increased the overall price level (0.33%) and reduced welfare by about 1.8%.

Collectively, this research shows that trade policy shocks are not isolated events but part of a wider set of geoeconomic instruments that shape firm performance, financial flows, and the structure of global value chains.

This research contributes to the literature on how financial intermediaries serve as a key channel through which central bank policies, including QE and QT, are transmitted to real

economic outcomes. He and Krishnamurthy (2013) develop a model showing that interventions such as lowering borrowing costs, purchasing assets, or injecting equity influence asset markets through the balance-sheet constraints of financial intermediaries, with equity capital injections being particularly effective because they directly relax the underlying capital scarcity. Di Maggio et al. (2020) provide empirical support for this mechanism, demonstrating that quantitative easing raises asset prices and injects liquidity into the financial system, thereby strengthening banks’ balance sheets, expanding their credit supply, and generating spillovers across financial markets and the broader economy. Cross-country evidence further indicates that monetary policy transmits not only through the traditional interest rate channel but also through intermediaries such as banks and dealers, whose balance-sheet capacity shapes the response of asset prices and credit supply, underscoring that the effectiveness of policy transmission depends critically on intermediary health (Baron and Muir, 2022). Consistent with this view, research on the deposits channel shows that when the Federal funds rate rises, banks—particularly those in concentrated deposit markets—transmit tighter monetary policy by contracting lending, with this mechanism fully accounting for the observed reduction in credit supply during monetary tightening episodes (Drechsler et al., 2017). Finally, Ray et al. (2024) highlight the role of institutional investors transmitting QE through their demand in Treasury auctions and segmented maturity markets, and ultimately lead to output and inflation.

Similarly, a related set of studies has examined how financial institutions, especially banks, respond to trade policy shocks. Alfaro et al. (2025) show that U.S. banks played a key role in supply-chain reallocation during the trade war by extending costly new loans to firms seeking alternative suppliers. Correa et al. (2023) find that trade policy uncertainty tightened U.S. bank lending, reducing borrowers’ debt growth and investment. Efung et al. (2023)) demonstrate that financial sanctions trigger immediate and coordinated responses by banks, underscoring the importance of financial channels in transmitting geopolitical risk. While these studies emphasize the role of banks, much less is known about how institutional

investors and financial intermediaries in equity markets adjust their portfolios in response to trade shocks, and how these reallocations shape firm outcomes. Calani et al. (2025) find trade-uncertainty shocks curb exports—especially cash-in-advance sales—while imports shift toward bank/CIA financing, prompting banks to reallocate credit toward larger importers without reducing total lending.

Despite these advances, an important gap remains. While researchers know tariff shocks affect directly exposed firms, shape global export reallocation, propagate through supply chains, and interact with financial frictions, people know little about how the U.S. financial intermediaries transmit tariff shocks through international equity markets. My paper contributes to filling this gap by combining tariff shocks, customs records, ownership data, and supply-chain linkages to examine how U.S. intermediaries reallocate capital in response to tariff shocks and how these reallocations amplify effects for both directly targeted and spillover firms.

3 Data

3.1 Tariff Shocks

In this paper, I construct a novel firm-level tariff-shock dataset, starting from the methodology of Fajgelbaum et al. (2020) and extending it using the timeline in Bown and Kolb (2022) and the USTR Lists 4A and 4B¹, which allows me to extend coverage through 2023.

Moreover, I incorporate additional sources of U.S. tariff shocks arising from antidumping (AD) and countervailing duty (CVD) measures. For this purpose, I use the AD and CVD data from the Temporary Trade Barrier Database (TTBD), which systematically documents temporary trade protection, including case initiation, investigation, outcomes, and the HS codes of affected products for each government’s AD/CVD actions. I focus on AD/CVD cases imposed by the United States and take the preliminary determination date—when

¹More details can be found on <https://ustr.gov/>

the authority announces whether tariffs will be imposed and at what tariff rates on specific products from the investigated countries—as the shock date. I further exclude cases that do not result in effective tariffs or that lack a recorded preliminary determination date. Due to data availability, I focus on post-2015 cases and on economies with sufficiently broad coverage: Australia, China, France, Germany, India, Italy, Japan, Mexico, Singapore, South Korea, Spain, Switzerland, Taiwan, and the United Kingdom.

3.2 U.S. Firm-level Custom Data

I restrict the my sample to publicly listed firms whose shares are held by the U.S. institutional investor and set the analysis window to begin in 2016, attributing to the relative sparsity of Panjiva shipments prior to that year. My core identification task is to determine which of these firms are exposed to U.S. tariff shocks. I define a firm as directly exposed if it exported the affected products to the United States in the two years preceding the tariffs’ announcement dates (Handley et al., 2025; Benguria et al., 2022). To implement this, I match (HS code, announcement date) pairs with shipment-level Panjiva records—customs (bill-of-lading) data linking importers and exporters across countries with dates, HS codes, product descriptions, quantities/weights, ports, and carriers—and flag firms meeting the pre-determined export criterion. I then match firm names from Panjiva to the FactSet fundamentals and ownership data using fuzzy-name algorithms (e.g., RapidFuzzy/FuzzyWuzzy), and cross-check manually about the quality of matching. Applying these procedures, I identify 394 directly exposed events.

3.3 Supply Chain Data

To capture indirect exposure along supply chains, I employ FactSet Revere relationships. These data record, for each link, the identities of customers, suppliers and competitors together with unique identifiers and the start and end dates of relationships. I retain supplier–customer–competitor links and construct a firm–pair–quarter panel.

3.4 Geographic Exposure Data

I use the FactSet Geographic Revenue Exposure dataset to construct firms' domestic sales exposure. These data report firms' sales breakdown across countries and regions, allowing me to measure the share of revenues generated in the United States. I define $Sales_{i,t}^d$ as firm i 's U.S. sales in year t and $Sales_{i,t}$ as its total sales. I use these variables to construct the domesticity measures, $Weighted\ Avg_{i,t}^{d,supplier}$ and $Weighted\ Avg_{i,t}^{d,customer}$, which capture how a firm's supply-chain partners are exposed to the U.S. market. These measures complement the simple domesticity indicators based on U.S. listing and headquarters location.

3.5 Ownership Data of the U.S. Institutional Investors

The FactSet Ownership database provides quarterly data on the equity holdings of U.S. institutional investors, including positions in both North American and non-North American firms. For the U.S. market, Figure 6 presents the domestic ownership structure of U.S. equities and motivates my focus on institutional investors. Although households and the foreign sector hold a large fraction of U.S. stocks, their portfolio holdings are difficult to observe. I therefore focus on four major institutional investor groups, mutual funds, pension funds, insurance companies, and hedge funds, whose equity positions are largely covered by FactSet ownership data. In Figure 7, my sample, for example at 2024 Q3, captures approximately 11.2 trillion dollars out of the 14.81 trillion dollars in aggregate holdings of these four groups, indicating that the data cover a substantial share of observable institutional ownership. This broad coverage provides a strong empirical foundation for studying how sophisticated investors rebalance portfolios in response to tariff shocks. My sample covers U.S. hedge funds, pension funds, insurance companies, and mutual funds over the period 2016–2023.

My core explanatory variable is the quarterly U.S. institutional investors' position-change ratio, which measures the proportional change in the ownership share of firm i from country j or U.S., or d , held by U.S. institutional investors between quarter $t - 1$ and quarter t .

Formally,

$$PCR_{i,t} = \frac{Position_{i,t} - Position_{i,t-1}}{Position_{i,t-1}}$$

This measure captures how actively investors adjust their holdings in response to shocks.

3.6 Quarterly Fundamental Data

All fundamental variables I use, including Tobin’s Q, R&D growth, capital expenditure growth, sales growth and net profit margin at the quarterly frequency, are also obtained from Compustat and FactSet Fundamentals (North America and International coverage). I exclude financial firms from the sample because their balance sheet structures and regulatory environments differ substantially from those of non-financial firms. I replace missing R&D values with 0, and to further address the influence of outliers. Most continuous variables—excluding event dummies—are winsorized at the 1% and 99% levels, except for variables with more extreme outliers, which are winsorized more aggressively. Moreover, all observations are converted into U.S. dollars to eliminate the influence of exchange-rate fluctuations, using exchange rate data from WRDS Macro Finance Society.

3.7 U.S. Daily Stock Return Data

To examine asset price responses to tariff shocks, I construct a daily stock-return panel for U.S. firms using Compustat North America. I merge daily returns with firm identifiers and compute abnormal returns relative to a standard factor model. Specifically, I obtain the Fama–French factors from Kenneth French’s data library and estimate abnormal returns, which are defined as the difference between realized firm returns and the model-implied expected returns, and I cumulate these abnormal returns over event windows around tariff shocks to form cumulative abnormal returns (CARs). This design allows me to track how U.S. spillover and competitor firms’ stock prices adjust in real time to tariff shocks,

complementing the quarterly ownership and fundamental analyses.

4 Empirical Designs

Using Panjiva shipment data and several sources for tariffs shocks, I construct two key event indicators to capture firms' exposure to tariff shocks. The first is the direct exposure dummy ($Exposure_{ijt}$), which equals one if firm i from country j exported any products subject to new tariffs during the two years prior to tariff imposition at time t . This variable identifies firms that are directly affected by U.S. tariff shocks through their export activities. The dummy equals one only in the tariff shock announcement year and zero otherwise, thus, identification comes from the information arrival at the tariff announcement, net of firms' subsequent adjustment dynamics in later years.

The second is the customer dummy ($Customer_{i,t}$), which equals one if firm i does not itself export any tariff-affected products to the United States in the two years preceding tariff imposition at time t , but is a downstream customer of at least one tariff-exposed firm, as identified using FactSet Revere supply-chain data. I define the supplier dummy ($Supplier_{i,t}$) analogously, identifying firms that are upstream suppliers of at least one tariff-exposed firm but are not directly exposed themselves. These variable capture indirect exposure transmitted along supply chains, allowing us to study how trade policy shocks propagate beyond directly targeted firms. I also define the *Competitor* indicator which is equal to one if a firm has a product-market competitor link with at least one tariff-exposed firm in time t based on FactSet Revere relationships data.

Importantly, because tariffs are imposed at the product level rather than at the country level, the resulting shock exposure is inherently heterogeneous across firms. Even within the same country, industry, or year, firms differ in whether they export the affected products, in the intensity of their reliance on those products, and in whether they are connected to exposed firms through supply-chain or product-market linkages. This micro-level variation is

central to the identification strategy. Unlike aggregate macroeconomic or monetary shocks, which tend to shift financing conditions, discount rates, or demand environments broadly across firms, tariff shocks generate differential treatment based on pre-existing product-level trade patterns and network positions. As a result, common macro disturbances may affect the overall level of firm outcomes, but they are unlikely to mechanically generate the cross-sectional exposure captured by *Exposure*, *Customer*, *Supplier*, and *Competitor*. Put differently, the identifying variation in my setting does not come from broad time-series movements in the economy, but from exogenous differences in firms’ predetermined exposure to newly tariffed products at the time of the policy announcement. This feature strengthens the interpretation of the estimated coefficients as capturing the causal effect of tariff shocks, rather than the confounding influence of contemporaneous macroeconomic conditions.

The empirical analysis against firms’ outcomes is conducted at the quarterly frequency, and I employ a fixed-effect regression framework to trace the dynamics of firm outcomes at these tariff shock events.

Moreover, I estimate panel regressions to examine whether U.S. institutional investors reduce or increase their positions in affected firms when tariff shocks occur. The results indicate that investors systematically reduce their holdings in exposed and spillover firms around tariff shocks, consistent with extensive evidence on selling-off in response to negative news (Christophe et al., 2004; Boehmer et al., 2008; Dechow et al., 2001; Christophe et al., 2010; Diether et al., 2009; Boehmer and Wu, 2013).

4.1 Effects of Tariff Shocks on U.S. Firms’ Cumulative Abnormal Returns

I begin with cumulative abnormal returns (CARs) as they provide a market-based measure of the valuation implications of tariff shocks. Shocks can reshape investors’ expectations about firms’ future cash flows and dividends, which are incorporated into stock prices. Moreover, the absence of pre-event abnormal returns provides a diagnostic for anticipation, supporting

the plausibility that tariff events are exogenous. Establishing these valuation responses motivates the subsequent analysis of real outcomes and performance.

I quantify the U.S.-stock-market price response to tariff-shocks using a standard two-step event-study design. In the first step, I estimate daily abnormal returns from the Fama–French three-factor model:

$$R_{i,t}^d - R_t^f = \beta_{Mi}(R_{Mt} - R_t^f) + \beta_{Si}SMB_t + \beta_{Hi}HML_t + \varepsilon_{i,t}^d,$$

where R_{idt} denotes firm i 's daily return (indexed by domestic market d and trading day t), R_t^f is the risk-free rate, and $(R_{Mt} - R_t^f, SMB_t, HML_t)$ are the Fama–French factors. The sample is restricted to firms that share the same two-digit SIC industry classification as the shocked firms.

$$AR_{i,t}^d \equiv \widehat{\varepsilon}_{i,t}^d.$$

In the second step, I aggregate abnormal returns over an event window to construct cumulative abnormal returns (CARs). For each firm–event pair, the CAR over window $[\tau_1, \tau_2]$ is

$$CAR_{i,t}^d[\tau_1, \tau_2] \equiv \sum_{t=\tau_1}^{\tau_2} AR_{i,t}^d.$$

I consider multiple event windows to capture both the immediate market reaction and potential delayed adjustment.

I then relate CARs to indicators for tariff-shock by estimating, for each event window,

$$CAR_{i,t}^d[\tau_1, \tau_2] = \gamma Shock_{i,t}^d + \mu_i + u_{i,t}^d,$$

where $Shock_{idt}$ is an indicator for the shocks of interest. In my setting, $Shock_{i,t}^d \in \{Supplier_{i,t}^d, Customer_{i,t}^d, Competitor_{i,t}^d\}$:

Consistent with the underlying economic mechanisms, I expect the sign of γ to vary

across transmission channels. For $Competitor_{idt}$, I expect $\gamma > 0$: tariff shocks can shift demand toward domestic rivals and soften product-market competition, generating a relative valuation gain for U.S. competitors. In contrast, for $Customer_{i,t}^d$ and $Supplier_{i,t}^d$, I expect $\gamma < 0$, as shocks transmitted through customer–supplier linkages can reduce revenues and thus expected future cash flows for upstream and downstream partners.

I further expect these effects to be more pronounced in longer event windows. Because the regressions capture indirect rather than direct exposure, relevant information may be incorporated into prices only gradually. Under slow-moving capital and inelastic demand for risky assets (Gabaix and Koijen, 2021; Duffie, 2010; Hou and Moskowitz, 2005), investor portfolio rebalancing and the propagation of shocks through supply chains may unfold with delay. As a result, short-window CARs may be attenuated, whereas longer-window CARs are more likely to capture the full valuation response.

4.2 U.S. Institutions’ Adjustment of Their Holdings in Response to U.S. Tariff Shocks.

Before examining the role of U.S. financial intermediaries in transmitting tariff shocks to firms, I first investigate how U.S. institutional investors adjust their holdings in response to U.S. tariff shocks. Specifically, I estimate the following two-way fixed effects (TWFE) panel regression:

$$PCR_{i,t}^j = \beta \cdot Tariff\ Shock_{i,t}^j + \Gamma X_{i,t-1}^j + \alpha_i + \alpha_t + \varepsilon_{i,t}^j.$$

Here, $PCR_{i,t}^j$ represents quarterly change ratios in U.S. institutional investors’ positions in firm i located in country j or the United States d at time t . A decline in PCR following tariff shocks would signal a capital withdrawal from the firm i . $Tariff\ Shock$ can be either $Exposure$, $Customer$, $Supplier$, $Competitor$. $Exposure$ is an indicator equal to one if a foreign firm is directly targeted by a U.S. tariff shock in quarter t . $Customer$ and $Supplier$

equal one if a firm is connected to at least one tariff-exposed firm through downstream customer or upstream supplier relationships in quarter t , respectively. *Competitor* is an indicator equal to one if a firm has a competitor relationship with at least one tariff-exposed firm in quarter t , based on FactSet Revere relationships data. The $X_{ij,t-1}$ here includes the lagged ROA, leverage ratio, returns and cash ratio.

4.3 Effects of U.S. Institutional Investors in Amplifying Tariff Shocks

The central research question of this paper is whether portfolio reallocation by U.S. institutional investors amplifies the effects of tariff shocks on firms. To test this hypothesis, I construct an interaction term between tariff shock indicators and changes in demeaned institutions' ownership shares. The moderating effect of institutions can be examined through the following regression specification:

$$\begin{aligned} Performance_{i,t}^j = & \sum_{k=-2}^0 \beta_k \cdot Tariff Shock_{i,t+k}^j + \sum_{k=-2}^0 \zeta_k \cdot PCR_{i,t+k}^j \\ & + \sum_{k=-2}^0 \gamma_k Tariff Shock_{i,t+k}^j \cdot PCR_{i,t+k}^j \\ & + \Gamma X_{i,t-1}^j + \alpha_i + \alpha_t + \varepsilon_{i,t}^j. \end{aligned}$$

Tariff Shock can be either *Exposure*, *Customer*, *Supplier*, *Competitor*. During tariff shock episodes, the U.S. institutions tend to reduce their equity holdings in affected firms to mitigate potential capital losses. Such sell-offs may exert downward pressure on stock returns and ultimately deteriorate firms' real fundamentals. I expect the coefficients γ_k and ζ_k to be positive if institutions' asset reallocations amplify the impact of tariff shocks on firm outcomes.

To capture the persistence of this amplification mechanism, I incorporate lagged terms

into the empirical specification, recognizing that firms’ real adjustments to financial shocks are unlikely to be immediate and may follow a lead–lag pattern. I also add the lagged stock returns as controls to solve the potential endogenous issues.

The asymmetric effects on exposed, spillover and competitor firms are consistent with recent findings in the trade and finance literature. Firms directly affected by tariff shocks often reallocate exports to alternative destinations, while increased domestic or third-market sales partially offset the contraction in their original export markets. Such substitution dynamics help stabilize aggregate sales and profitability, thereby attenuating the observable effects on overall sales growth and firm performance (Jiao et al., 2024; Jiang et al., 2023; Benguria and Saffie, 2024). Alternatively, due to tariff evasion and re-routing practices, Chinese exporters’ sales to the U.S. may have been partly preserved (Che et al., 2025). In contrast, both empirical and theoretical evidence suggests that the sales and other fundamentals of spillover firms—those indirectly affected through supply chain linkages or market dependencies—tend to deteriorate (Huang et al., 2023; Grossman et al., 2024). Moreover, Fajgelbaum et al. (2020) show that U.S. tariffs partially diverted demand toward untargeted countries, implying that competitor firms may gain market share. These evidences motivate me to further investigate the mechanisms.

5 Empirical Results

5.1 Descriptive Statistics

Table 1 presents summary statistics for the key variables in my analysis, separately for foreign and U.S. firms. The foreign-firm panel contains 175,654 firm-quarter observations for *PCR*, while the U.S.-firm panel contains 81,908 observations. In both samples, *PCR*—the quarterly change ratio in U.S. institutional investors’ positions—is centered close to zero but exhibits substantial dispersion. For foreign firms, the mean PCR is 0.021 with a median of 0.000 and an interquartile range from 0.000 to 0.037; for U.S. firms, the mean is 0.028

with a median of 0.010 and a wider upper tail. The positive mean also indicates an overall expansion in U.S. institutions' holdings, consistent with the pattern shown in Figure 3

Foreign firms exhibit moderate growth and valuation dynamics, together with pronounced skewness in several fundamentals. Quarterly sales growth averages 1.6% (median 2.3%), and quarterly R&D growth averages 1.9% (median 2.0%), while Tobin's Q averages 1.77 with a median of 1.12, indicating valuations generally above book values. Profitability is positive but modest: net margins average 0.6% with a median of 0.1%, and ROA averages 1.13% with a median of 0.13%, consistent with a right-skewed distribution. Balance-sheet measures are also skewed: the leverage ratio averages 27.35% but has a median of 2.52%, and cash holdings average 25.14% with a median cash-to-assets ratio of 2.65%. U.S. firms display different distributional patterns. Quarterly sales growth is higher on average at 4.8% (median 2.3%), and Tobin's Q is higher at 1.94 (median 1.42). Profitability outcomes are substantially more dispersed: net margins average -2.56 with a median of 2.1%, and mean ROA is -0.032 with a median of 0.27%, consistent with a sizable left tail of loss observations. Investment measures show weaker average capex growth (mean -0.8% , median 1.9%) and R&D growth of 1.0% (median 2.6%). Overall, the table highlights substantial heterogeneity in both firm fundamentals and U.S. institutions' position changes, motivating the fixed-effects specifications used in the subsequent analysis.

5.2 Cumulative Abnormal Returns for U.S. Spillover and Competitor Firms

Figure 8 plots event-time coefficients for cumulative abnormal returns (CARs) of U.S. competitor firms around U.S. tariff shocks. In the pre-shock period, the estimated CARs are economically small and statistically indistinguishable from zero, with confidence intervals generally overlapping the zero line. This absence of pre-trends suggests that the tariff shocks are not systematically anticipated by the market and that competitor firms do not experience differential price run-ups prior to the tariff announcement.

Following the tariff shock (event time $t \geq 0$), CARs increase steadily and become persistently positive. The post-shock coefficients rise over the event horizon, indicating a sustained appreciation in competitors' stock prices rather than a transitory announcement effect. And in another word, this pattern implies that U.S. tariff shocks are largely unanticipated and generate positive valuation effects for U.S. competitors, consistent with demand reallocation and improved competitive positioning in product markets after the shock.

Figure 9 shows the event-time path of CARs for U.S. customer firms around U.S. tariff shocks. Pre-shock coefficients fluctuate around zero and are statistically indistinguishable from zero, indicating no evidence of differential price run-ups prior to the shock. After the shock, however, CARs gradually turn negative and continue to drift downward, becoming persistently below zero in longer post-shock windows. Thus, the absence of pre-trends supports the interpretation that tariff shocks are largely unanticipated, while the delayed post-shock decline indicates that U.S. customer firms incur valuation losses following tariff shocks, consistent with negative propagation through customer relationships and gradual incorporation of supply-chain disruptions into stock prices.

Figure 10 presents the CAR dynamics for U.S. supplier firms around U.S. tariff shocks. Similarly, the estimates show little movement before the event, with confidence intervals generally covering zero, which suggests that supplier firms do not experience abnormal returns prior to the shock. In the short post-shock windows, the response remains limited, but the coefficients begin to fall more clearly after longer horizons. By the end of the event window, CARs are substantially negative, indicating that tariff shocks impose valuation losses on U.S. suppliers with a delay.

5.3 Role of U.S. Institutional Investors in Shock Transmission

Beyond the price effect, Table 2 reports quarterly panel regressions examining how U.S. institutional investors rebalance their equity positions in response to U.S. tariff shocks, using the U.S. institutional investors' position change ratio (PCR) as the dependent variable. I

begin with the U.S.-firm sample. Column (1) shows that U.S. customer firms experience a statistically significant decline in institutional ownership following tariff shocks. The coefficient of -0.02607 implies that, in shock quarters, U.S. institutions reduce their positions in U.S. customer firms by approximately 2.6 percentage points. Column (2) documents an even larger response for U.S. supplier firms: the coefficient of -0.03028 indicates a 3.0 percentage point decline in *PCR*. These estimates suggest that institutional investors respond to tariff-induced supply-chain risk by withdrawing capital from domestically listed firms that are economically linked to tariff-exposed firms through customer-supplier relationships.

In contrast, the ownership response for U.S. competitor firms is positive but less robust. Column (3) shows that the coefficient on *Competitor* is 0.0210 and statistically significant, implying that U.S. institutions increase their positions in domestic rival firms by about 2.1 percentage points after tariff shocks. This finding is consistent with a portfolio reallocation channel in which institutions rotate capital toward firms expected to benefit from weakened foreign competitors and improved product-market conditions. And once year-quarter fixed effects are included in column (4), the coefficient declines to 0.00789 and becomes statistically insignificant.

I then turn to the foreign-firm sample. Column (5) shows that directly exposed foreign firms experience a statistically significant decline in *PCR*. The coefficient of -0.03496 implies that U.S. institutions reduce their positions in tariff-exposed foreign firms by approximately 3.5 percentage points in shock quarters. This result indicates that U.S. institutions actively divest from foreign firms directly targeted by U.S. trade policy shocks. Column (6) further shows that foreign customer firms connected to exposed firms experience a -0.02725 decline in *PCR*, suggesting that institutional selling extends beyond directly targeted firms to firms indirectly affected through foreign supply-chain linkages. By contrast, column (7) shows that foreign competitor firms experience a positive and statistically significant increase in *PCR* of 1.24 percentage points, consistent with capital reallocation toward firms that may gain from tariff-induced shifts in competitive conditions.

Table 2 provides evidence that U.S. institutional investors do not passively absorb tariff shocks. Instead, they actively rebalance their portfolios in a direction consistent with the underlying economic exposure of firms. Institutions sell firms that are directly exposed to tariffs or indirectly exposed through customer–supplier linkages, while reallocating capital toward competitor firms that are more likely to benefit from the disruption. The cross-sectional asymmetry in institutional trading is unlikely to reflect passive benchmark rebalancing alone, but instead points to active portfolio reallocation based on investors’ assessment of firm-level exposure to tariff-induced downside risk and competitive gains.

Table 3 examines U.S. customer firms, defined as domestic firms that are connected as customers to tariff-exposed foreign firms, and shows that U.S. institutional trading has economically important implications for these firms’ fundamentals. Across columns, the coefficients on $PCR_{i,t}^d$ and its lags are generally positive for real activity, indicating that lower institutional ownership growth is associated with weaker sales growth and lower investment. For example, in column (1), the estimate on $PCR_{i,t}^d$ is 0.0166, implying that a one-standard-deviation decline in PCR (0.246) reduces quarterly sales growth by about $0.0166 \times 0.246 \approx 0.0041$, or 0.41 percentage points, holding fixed firm and year-quarter effects and firm-level controls.

More importantly for the mechanism, several customer interaction terms are positive and statistically meaningful, indicating that institutional selling during tariff-shock quarters amplifies the deterioration in customer firms’ real outcomes. In column (1), $Customer \times PCR_{i,t}^d$ is 0.0899, so a one-standard-deviation decline in PCR during customer-shock quarters translates into an additional $0.0899 \times 0.246 \approx 0.022$, or 2.2 percentage points, reduction in sales growth relative to non-customer firms. Similarly, in column (3), $Customer \times PCR_{i,t}^d$ equals 0.0976, implying an additional $0.0976 \times 0.246 \approx 0.024$, or 2.4 percentage points, contraction in R&D growth when institutions cut positions in customer firms. The longer-horizon interactions are also sizable: $Customer \times PCR_{i,t-2}^d$ is 0.108 for R&D growth and 0.146 for capex growth in columns (3)–(4), so a one-standard-deviation sell-off maps into about 0.0266 and

0.0359, or 2.7 and 3.6 percentage points, larger long-run declines in R&D and capex growth two quarters later, respectively.

Given that U.S. institutions reduce holdings of U.S. customer firms around tariff shocks, the positive interaction coefficients imply that customer firms experience worse fundamentals precisely when institutional capital withdraws. This pattern is consistent with a financial-amplification channel: tariff shocks first weaken firms through their customer linkages to exposed foreign firms, and institutional selling further tightens these firms' capital-market conditions, amplifying the decline in real activity. Finally, the customer indicator itself points to delayed operating stress: $Customer_{i,t-1}^d$ is negative and significant in the net margin regression, consistent with supply-chain disruptions and cost pass-through materializing with a lag even when contemporaneous revenue effects are muted.

Table 4 examines U.S. supplier firms, defined as domestic firms that supply to tariff-exposed foreign firms. The results show that U.S. institutional trading is positively associated with supplier firms' real activity and valuation. Across columns, the coefficients on $PCR_{i,t}^d$ and its lags are mostly positive and statistically significant. For example, in column (1), the estimate on $PCR_{i,t}^d$ is 0.0212, implying that a one-standard-deviation decline in PCR (0.246) is associated with a $0.0212 \times 0.246 \approx 0.0052$, or 0.52 percentage point, reduction in quarterly sales growth. The lagged PCR coefficients are also positive for sales growth, profitability, R&D growth, and Tobin's Q, suggesting that institutional capital withdrawal is followed by weaker fundamentals and lower valuation among U.S. suppliers.

The supplier interaction terms provide more direct evidence on the amplification channel, although the effects are concentrated in profitability and valuation rather than sales or R&D. In column (2), $Supplier \times PCR_{i,t}^d$ is 0.0572, implying that a one-standard-deviation institutional sell-off during supplier-shock quarters leads to an additional $0.0572 \times 0.246 \approx 0.0141$, or 1.4 percentage point, decline in net profit margin relative to non-supplier firms. In column (4), the longer-horizon interactions are positive and statistically meaningful: $Supplier \times PCR_{i,t-1}^d$ equals 0.172, and $Supplier \times PCR_{i,t-2}^d$ equals 0.308. These estimates

imply that a one-standard-deviation decline in PCR maps into additional reductions in Tobin’s Q of about 0.042 and 0.076 one and two quarters later, respectively. Given that U.S. institutions reduce holdings of U.S. supplier firms around tariff shocks, these positive interaction coefficients suggest that institutional selling amplifies the decline in supplier firms’ profitability and valuation.

Finally, the supplier indicator itself points to direct valuation pressure. $Supplier_{i,t}^d$ is negative and significant in the Tobin’s Q regression, indicating that U.S. suppliers experience an immediate valuation discount when they are linked to tariff-exposed firms. Overall, the evidence suggests that tariff shocks affect U.S. suppliers mainly through profitability and valuation channels, and that institutional capital withdrawal strengthens these adverse effects over subsequent quarters.

Table 5 turns to U.S. competitor firms, domestic firms that compete with at least one tariff-exposed firm, and examines how U.S. institutional trading affects their real outcomes around tariff shocks. The coefficients on $PCR_{i,t}^d$ are positive in the sales regressions, indicating that net selling-off (lower PCR) coincides with slower sales growth. In column (1), the coefficient on $PCR_{i,t}^d$ is 0.0155, so a one-standard-deviation increase in PCR (0.246) corresponds to a $0.0155 \times 0.246 \approx 0.0038$ increase in quarterly sales growth (0.38 percentage points), holding fixed firm and year-quarter effects and controls. The lagged PCR terms are also positive for sales growth, R&D growth, and capex growth, consistent with institutional rebalancing translating into persistent changes in real activity.

The competitor indicator itself suggests that tariff shocks reallocate product-market conditions in favor of domestic rivals. In particular, $Competitor_{i,t}^d$ is positive in columns (1)–(2): competitor firms experience 2.31 percentage points higher contemporaneous sales growth and 5.43 percentage points higher net margins if they are linked to exposed rivals. Economically, these estimates are consistent with tariff shocks weakening exposed firms and allowing competitors to expand demand and improve pricing power or markups, at least in the short run.

More importantly for the mechanism, several interaction terms between competitor status and institutional trading are positive and economically meaningful, implying that institutional trading can amplify the extent to which competitor firms benefit from tariff shocks. In column (2), $Competitor \times PCR_{i,t-1}^d$ equals 0.278, so a one-standard-deviation increase in PCR in the prior shock quarter implies a $0.278 \times 0.246 \approx 0.068$ (6.8 percentage points) higher net margin for competitor firms in the current quarter, relative to non-competitors. In column (4), $Competitor \times PCR_{i,t-2}^d$ is 0.226, implying that a one-standard-deviation increase in PCR at the tariff shock two quarters earlier leads to about a $0.226 \times 0.246 \approx 0.056$ (5.6 percentage points) higher CapEx growth for competitor firms in the current quarter. These patterns are consistent with heterogeneous institutional responses in which some investors rotate toward domestic rivals following tariff shocks, and such capital reallocation is followed by stronger profitability and investment among competitor firms.

The evidence shows that U.S. tariff shocks have heterogeneous effects on domestic firms. U.S. customers and suppliers connected to tariff-exposed foreign firms tend to experience weaker fundamentals, including slower sales growth, lower investment, and worse operating performance, particularly when institutional investors reduce holdings. By contrast, U.S. competitor firms generally benefit from these shocks, with higher profitability and stronger capital expenditures, consistent with gains in market share and improved product-market conditions. More importantly, they indicate that the institutional investor channel transmits tariff-related risk beyond supply chains, making investor capital reallocation an important mechanism in the broader transmission of trade shocks. This mechanism should therefore be taken into account in future research on the real effects of trade or monetary policy.

Table 6 reports quarterly results for foreign firms that are directly exposed to U.S. tariffs, relative to other foreign firms, and shows that tariff exposure leads to a sizable and persistent valuation loss. Tobin's Q falls by 0.200 in the shock quarter and remains negative in the subsequent quarters (0.149 at $t - 1$ and 0.093 at $t - 2$), indicating that markets revise down exposed firms' prospects beyond an immediate announcement effect. Consistent with insti-

tutional sell-offs around tariff shocks, the interaction term $Exposure_{i,t}^j \times PCR_{i,t}^j$ is strongly positive for R&D growth (0.476), implying that larger ownership reductions at the time of exposure translate into a sharper innovation contraction among exposed firms. Economically, a one-standard-deviation decline in foreign-firm PCR (0.20) maps into an additional $0.476 \times 0.20 \approx 0.095$ (9.5 percentage points) reduction in quarterly R&D growth for exposed firms relative to non-exposed firms, supporting the hypothesis that capital withdrawal amplifies the real consequences of exposure along the innovation margin.

Table 7 focuses on foreign customer firms, which are indirectly linked to tariff-exposed firms through customer relationships. Customer exposure leads to valuation declines: Tobin's Q falls by 0.0575 in the shock quarter and remains negative at $t - 2$ (0.0327). Institutional trading amplifies these effects with a lag. The interaction $Customer_{ijt} \times PCR_{i,t-1}^j$ is positive for Tobin's Q (0.153), implying that a one-standard-deviation decline in PCR (0.20) generates an additional $0.153 \times 0.20 \approx 0.031$ reduction in Tobin's Q for customer firms relative to non-customer firms. Customer effects on innovation also emerge with delay, as $Customer_{i,t-2}^j$ is negative for R&D growth (-0.0428), consistent with gradual propagation of supply-chain disruptions into longer-horizon investment decisions.

Table 8 examines foreign competitor firms, defined relative to foreign non-competitors. Here, the institutional channel operates in the opposite direction. While the competitor indicator itself shows limited average effects on fundamentals, the interaction $Competitor_{ijt} \times PCR_{i,t}^j$ is strongly positive for Tobin's Q (0.652), indicating that competitor firms experience larger valuation gains when U.S. institutions increase their holdings. Economically, a one-standard-deviation increase in PCR (0.20) corresponds to a $0.652 \times 0.20 \approx 0.130$ increase in Tobin's Q for competitor firms relative to non-competitors, consistent with heterogeneous trading responses in which some institutions rotate toward potential winners of demand reallocation following tariff shocks.

U.S. tariff shocks generate broad negative effects for domestic customer and supplier firms linked to exposed firms through supply-chain networks, with the adverse impact visible in

both market-based and real-side outcomes. Although these firms are not directly subject to the tariff, they are indirectly harmed when exposed firms cut purchases, scale back production, or reorganize sourcing, thereby transmitting the shock along customer-supplier linkages. At the same time, tariffs create relative winners among certain U.S. competitors by reallocating product-market demand away from foreign firms and toward domestic substitutes, allowing these firms to increase sales, improve profitability, and expand investment. While exposed firms are directly harmed by higher trade barriers, foreign spillover firms may suffer even larger losses because the shock propagates through customer-supplier linkages and is amplified by institutional investor channels. As exposed firms cut orders, production, or sourcing, foreign upstream and downstream partners face declines in sales and tighter liquidity. Because cross-border relationships are costly to replace and inputs or buyers are often not easily substitutable, these spillover effects can be more severe and persistent than the direct effect on exposed firms. By contrast, gains for competitor firms are neither universal nor guaranteed. In many cases, the fragmentation of global supply chains constrains their ability to scale production efficiently, limiting the extent to which they can benefit from protection-induced demand reallocation. As a result, tariff shocks generate a strongly asymmetric pattern: they systematically depress outcomes for exposed and spillover firms, especially foreign spillovers, while only selectively benefiting U.S. competitors.

In terms of economic magnitudes, Tables 3–5 and Tables 6–8 suggest that the institutional channel is quantitatively stronger for U.S. firms than for foreign firms when evaluated on comparable outcomes. Using a one-standard-deviation benchmark for PCR (0.246 for U.S. firms and 0.20 for foreign firms), the customer amplification on R&D growth is larger in the U.S. sample: for example, the coefficient of $Customer_{i,t-2}^j \times PCR_{i,t-2}^j$ is 0.108, implies $0.108 \times 0.246 \approx 0.027$, or 2.7 percentage points for U.S. spillover firms (Table 3), whereas $Customer_{i,t-2}^j \times PCR_{i,t-2}^j = 0.0732$ implies $0.0878 \times 0.20 \approx 0.018$, 1.8 percentage points, for foreign customer firms (Table 7). Overall, these comparisons indicate that institutional trading transmits tariff shocks more strongly within the U.S. connected-firm network than

among foreign connected firms.

Within the foreign-firm sample, direct exposure effects also dominate spillover effects along the same margins. For example, in terms of innovation, the amplification is materially stronger under direct exposure: a one-standard-deviation decline in *PCR* implies an additional $0.476 \times 0.20 \approx 0.095$ reduction in quarterly R&D growth for exposed firms, whereas the corresponding spillover amplification on R&D growth is smaller. This evidence suggests that direct tariff exposure generates the sharpest real adjustments, while spillover effects remain economically meaningful but attenuated.

6 Further Evidence on Supply-Chain Reorganization

Having documented that tariff shocks affect firms' valuation and real outcomes through both supply-chain linkages and institutional investor rebalancing, I next examine whether these shocks also reshape firms' international operations and supply-chain structure. In particular, if tariff shocks disrupt existing cross-border production networks, affected U.S. firms may respond not only by adjusting investment, but also by reorganizing their supply chains, reallocating sales toward domestic markets, and selecting more domestic-oriented supply-chain partners. Therefore, I analyze both firms' foreign sales growth and changes in their supply-chain partner composition to test whether tariff shocks lead to a contraction in foreign-market activity and a shift toward domestic or more domestically oriented partners, consistent with supply-chain reconstruction and increasing domesticity after the trade shock.

6.1 Tariff Shocks Depress Foreign Sales Growth

To examine whether tariff shocks also affect firms' foreign-market performance, I estimate the following annual specification using firms' foreign sales growth as the dependent variable:

$$ForeignSalesGrowth_{i,t}^d = \beta_0 Shock_{i,t}^d + \beta_1 Shock_{i,t-1}^d + \Gamma X_{i,t-1}^d + \alpha_i + \alpha_t + \varepsilon_{i,t}^d,$$

where *Shock* is either *Supplier* or *Customer*, depending on whether a U.S. firm is connected to at least one tariff-exposed firm as a supplier or customer. In alternative specifications, I replace firm and year fixed effects with sector-by-year fixed effects to absorb time-varying shocks common to firms within the same sector.

Table 9 reports annual regression results for the effect of U.S. tariff shocks on firms' foreign sales growth. The results show that supply-chain exposure to tariff shocks is followed by weaker foreign-market performance. For U.S. supplier firms, the contemporaneous coefficients are negative but statistically insignificant, while the lagged supplier indicator is negative and significant across specifications. In column (1), $Supplier_{i,t-1}^d$ equals -6.814 , implying that supplier firms experience a 6.8 percentage point decline in foreign sales growth in the year after the shock. The effect remains negative and statistically significant after including firm and year fixed effects in column (2), with a coefficient of -4.675 . These estimates suggest that the adverse effect on suppliers materializes with a lag, consistent with the gradual adjustment of foreign sales contracts and supply-chain relationships.

The customer-firm results show a similar delayed pattern. The contemporaneous customer coefficients are small and statistically insignificant, but $Customer_{i,t-1}^d$ is negative and significant in both specifications. Column (3) shows that customer firms experience a 4.5 percentage point decline in foreign sales growth in the year after the shock, and column (4) shows that the effect remains economically meaningful at 3.3 percentage points after controlling for firm and year fixed effects. Overall, the results indicate that tariff shocks weaken firms' foreign sales growth through both supplier and customer linkages, but the effects appear mainly with a one-year lag rather than immediately in the shock year. This pattern is consistent with supply-chain disruptions gradually spilling over into firms' foreign-market operations as firms adjust sourcing, production, and customer relationships after the tariff shock.

6.2 Domesticity as a Buffer against Trade-War Shocks

I construct two measures to capture the domesticity of supplier firm i 's customer network. The first measure is based on the share of firm i 's customers that are U.S.-listed and headquartered in the United States:

$$Simple\ Avg_{i,t}^{d,supplier} = \frac{1}{N_{i,t}^C} \sum_{c \in \mathcal{C}_{i,t}} D_{c,t}^d.$$

The second measure is based on the U.S. sales share of firm i 's customer base:

$$Weighted\ Avg_{i,t}^{d,supplier} = \frac{\sum_{c \in \mathcal{C}_{i,t}} Sales_{c,t}^d}{\sum_{c \in \mathcal{C}_{i,t}} Sales_{c,t}}.$$

Here, $\mathcal{C}_{i,t}$ denotes the set of customers of supplier firm i , and $N_{i,t}^C$ is the number of these customers. $D_{c,t}^d$ equals one if customer c is U.S.-listed and headquartered in the United States, and zero otherwise. $Sales_{c,t}^d$ denotes customer c 's sales generated in the United States, while $Sales_{c,t}$ denotes its total sales. Thus, the first measure captures the domestic composition of supplier i 's customer network, whereas the second measure captures the domestic sales exposure of supplier i 's customer base.

Similarly, I construct analogous domesticity measures for U.S. customer firms, denoted as $Simple\ Avg_{i,t}^{d,customer}$ and $Weighted\ Avg_{i,t}^{d,customer}$, to capture the domestic composition and domestic sales exposure of customer firm i 's supplier network.

I then define TW_{t-1} as an indicator for the first U.S.–China trade war period:

$$TW_{t-1} = \mathbf{1}\{2018Q1 \leq t-1 \leq 2019Q4\}.$$

TW_{t-1} equals one if the previous quarter falls within the first U.S.–China trade war period, from 2018Q1 to 2019Q4, and zero otherwise.

Table 10 examines whether domestic supplier-network orientation mitigates the adverse effect of the first U.S.–China trade war on U.S. firms' performance. The coefficient on

TW_{t-1} is negative and statistically significant across all columns, indicating that the trade war weakens firm performance: sales growth, profitability, and Tobin’s Q all decline during the trade-war period. However, the interaction terms with domestic supplier measures are positive across specifications, suggesting that firms with more domestically oriented supplier networks suffer smaller losses. In columns (1)–(3), the interaction with $Simple\ Avg_{i,t-1}^{d,supplier}$ is positive for sales growth, profitability, and Tobin’s Q , and is statistically significant for Tobin’s Q , implying that firms whose suppliers are more connected to domestic customers experience less severe valuation losses. Similarly, in columns (4)–(6), the interaction with $Weighted\ Avg_{i,t-1}^{d,supplier}$ is positive across all outcomes and statistically significant for sales growth, indicating that firms connected to suppliers with greater U.S. sales exposure experience a smaller decline in sales growth during the trade war. Although some interaction coefficients are not statistically significant, their consistently positive signs suggest that domestic supplier orientation mitigates, rather than amplifies, the negative performance effects of the trade war. Overall, the results indicate that the trade war depresses supplier firms’ performance, but firms with more domestic supplier networks are partially insulated from these losses.

Table 11 examines whether domestic customer-network orientation mitigates the adverse effect of the first U.S.–China trade war on U.S. firms’ performance. The coefficient on TW_{t-1} is negative and statistically significant across all columns, indicating that the trade war reduces sales growth, profitability, and Tobin’s Q . However, the interaction terms with domestic customer measures are consistently positive, suggesting that firms with more domestically oriented customer networks suffer smaller losses during the trade-war period. In columns (1)–(3), the interaction with $Simple\ Avg_{i,t-1}^{d,customer}$ is positive for all outcomes and statistically significant for sales growth and Tobin’s Q , implying that firms whose customers are more domestic experience less severe declines in operating performance and valuation. Similarly, in columns (4)–(6), the interaction with $Weighted\ Avg_{i,t-1}^{d,customer}$ is positive across outcomes and strongly significant for Tobin’s Q , indicating that firms connected to customers

with greater U.S. sales exposure face smaller valuation losses. Although some interaction coefficients are statistically insignificant, their positive signs are consistent with a mitigating role of domestic customer orientation. Overall, the evidence suggests that the trade war weakens firm performance, but firms with more domestic customer networks are partially insulated from these adverse effects.

6.3 Tariff Shocks Lead to Domestic Supply-Chain Reorganization

Table 12 examines whether U.S. tariff shocks induce supply-chain reorganization toward more domestic partners. The explanatory variables, $Supplier_{i,t}^d$ and $Customer_{i,t}^d$, capture whether firm i is connected through the supply chain to at least one tariff-exposed firm as a supplier or customer. The results show that, following tariff shocks, both supplier and customer firms become more domestically oriented in their supply-chain networks. In columns (1)–(2), the coefficients on $Supplier_{i,t}^d$ and $Supplier_{i,t-1}^d$ are positive and statistically significant, indicating that U.S. supplier firms connected to exposed firms are more likely to have supplier networks that are headquartered in the United States or have greater U.S. sales exposure. In columns (3)–(4), the customer-side results display a similar pattern. The coefficients on $Customer_{i,t}^d$ and $Customer_{i,t-1}^d$ are positive and significant for the simple-average measure, suggesting that U.S. customer firms also shift toward more domestic partners after tariff shocks. Although the weighted-average customer domesticity coefficients in column (4) are statistically insignificant, their signs are mostly positive, consistent with the same direction of adjustment. Since supplier–customer contracts are often determined before tariff shocks occur and many existing relationships continue to be implemented afterward, detecting such positive changes is economically meaningful. In short, the evidence suggests that tariff shocks not only affect firms’ performance, but also induce U.S. firms to reorganize their supply chains toward U.S. partners or partners with more U.S.-sales-dominant operations.

Table 13 examines whether the first U.S.–China trade war is associated with supply-chain reorganization toward more domestic partners, using TW_{t-1} as the main regressor.

The results show that U.S. firms become more domestically oriented during the trade-war period. In columns (1)–(2), the coefficients on TW_{t-1} are positive for both supplier domesticity measures, and the effect is statistically significant for the weighted-average measure. This indicates that, during the trade war, firms are increasingly connected to suppliers whose sales are more concentrated in the United States. In columns (3)–(4), the coefficients are positive and statistically significant for both customer domesticity measures, suggesting that firms’ customer networks also become more domestic, both in terms of customers’ headquarters location and U.S. sales exposure. Given that supplier–customer relationships are often predetermined before trade shocks occur and many existing contracts continue to be implemented after the shock, the positive estimates are economically meaningful. Overall, the evidence suggests that the U.S.–China trade war induces firms to reorganize their supply chains toward U.S. partners or partners with more U.S.-sales-dominant operations.

7 Conclusion

This paper studies how U.S. institutional investors transmit and amplify the effects of U.S. tariff shocks on asset prices, firm performance, and supply-chain reorganization. Using detailed product-level tariff data combined with customs shipment records from Panjiva, I construct firm-level measures of tariff incidence that capture both direct exposure to U.S. tariff actions and indirect exposure through supply-chain linkages. I classify firms into directly targeted firms (*exposed*), firms indirectly affected through customer or supplier relationships (*customer* and *supplier*), and firms connected through product-market rivalry (*competitor*). Merging these exposure measures with firm fundamentals, supply-chain relationships, and institutional holdings, I quantify how institutional rebalancing shapes the cross-firm propagation of tariff shocks.

Asset-price evidence indicates that tariff shocks are largely unanticipated by financial markets. Prior to tariff announcements, cumulative abnormal returns (CARs) for U.S. cus-

tomer, supplier, and competitor firms are statistically indistinguishable from zero. Following the shock, however, CARs diverge across exposure types: U.S. customer and supplier firms experience negative CARs over longer event windows, whereas U.S. competitor firms exhibit persistently positive CARs. These asymmetric valuation responses are consistent with tariff shocks propagating through customer-supplier linkages while reallocating demand toward competitors.

Turning to firm performance, tariff shocks generate economically meaningful real effects, with strong heterogeneity across exposure types. For foreign firms, direct exposure produces the most severe adjustments, including a large and persistent decline in Tobin’s Q , and sharp contractions in innovation when institutional investors withdraw capital. Foreign customer effects are smaller but remain economically meaningful and often materialize with a lag, consistent with gradual pass-through of supply-chain disruptions. For U.S. connected firms, the real effects differ across supply-chain positions: U.S. customer firms experience weaker sales growth, R&D growth, and capital expenditure, while U.S. supplier firms exhibit declines mainly in profitability and valuation. By contrast, U.S. competitor firms experience improved profitability and investment, consistent with demand reallocation and strengthened product-market conditions.

A central contribution of the paper is to identify a financial amplification channel operating through U.S. institutional investors. In response to tariff shocks, U.S. institutions reduce their holdings in exposed, customer, and supplier firms, as reflected in significant declines in the position-change ratio (PCR), while increasing holdings of some competitor firms. This asymmetric trading pattern is difficult to reconcile with passive benchmark rebalancing alone and instead points to active portfolio reallocation based on firm-level exposure to tariff-induced downside risk and competitive gains. Interaction specifications show that when PCR falls during shock periods, the subsequent deterioration in firm fundamentals becomes substantially more pronounced: institutional sell-offs amplify contractions in sales growth, innovation, investment, profitability, and valuation among exposed and supply-

chain-connected firms. In contrast, for competitor firms, increases in institutional holdings magnify gains in profitability and investment, consistent with capital rotating toward potential winners of tariff-induced demand reallocation. Importantly, the implied magnitudes of these amplification effects are larger for U.S. connected firms than for comparable foreign connected firms, indicating that the institutional channel is particularly powerful within domestic networks.

Finally, I provide further evidence that tariff shocks reshape firms' operations and supply-chain structure. U.S. firms connected to tariff-exposed firms experience weaker foreign sales growth, suggesting a contraction in foreign-market activity through both customer and supplier linkages. At the same time, firms with more domestically oriented supplier or customer networks suffer smaller declines in performance during the trade-war period. I also find that tariff shocks induce firms to shift toward U.S. partners or partners with more U.S.-sales-dominant operations. Overall, the findings imply that U.S. tariff shocks generate far-reaching consequences through the interaction of supply-chain linkages, capital-market reallocations, and supply-chain reconstruction. By documenting these mechanisms, the paper contributes to the intersection of international trade, corporate finance, and institutional investors, and highlights the importance of jointly accounting for production networks, investor behavior, and firms' real-side adjustment when assessing the effects of trade policy.

References

- Ahn, D. P. and Ludema, R. D. (2020). The sword and the shield: The economics of targeted sanctions. *European Economic Review*, 130:103587.
- Alfaro, L., Brussevich, M., Minoiu, C., and Presbitero, A. F. (2025). Bank financing of global supply chains. Technical report, National Bureau of Economic Research.
- Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *The quarterly journal of economics*, 131(4):1593–1636.

- Baron, M. and Muir, T. (2022). Intermediaries and asset prices: International evidence since 1870. *The Review of Financial Studies*, 35(5):2144–2189.
- Barrot, J.-N. and Sauvagnat, J. (2016). Input specificity and the propagation of idiosyncratic shocks in production networks. *The Quarterly Journal of Economics*, 131(3):1543–1592.
- Benguria, F., Choi, J., Swenson, D. L., and Xu, M. J. (2022). Anxiety or pain? the impact of tariffs and uncertainty on chinese firms in the trade war. *Journal of International Economics*, 137:103608.
- Benguria, F. and Saffie, F. (2024). Escaping the trade war: Finance and relational supply chains in the adjustment to trade policy shocks. *Journal of International Economics*, 152:103987.
- Boehmer, E., Jones, C. M., and Zhang, X. (2008). Which shorts are informed? *The Journal of Finance*, 63(2):491–527.
- Boehmer, E. and Wu, J. (2013). Short selling and the price discovery process. *The Review of Financial Studies*, 26(2):287–322.
- Bown, C. P. (2014). Temporary trade barriers database: Update through 2013. *World Bank*. Available at: <http://econ.worldbank.org/ttbd/>, June.
- Bown, C. P. and Kolb, M. (2022). Trump’s trade war timeline: An up-to-date guide.
- Calani, M., Margaretic, P., and Moreno, D. (2025). Navigating trade uncertainty: The role of trade financing and the spillover effects. *Journal of International Economics*, 153:104043.
- Caldara, D. and Iacoviello, M. (2022). Measuring geopolitical risk. *American economic review*, 112(4):1194–1225.
- Camous, A. and Cooper, R. (2021). Political activism and the provision of dynamic incentives: Growing the pie in the battle for redistribution. *The Economic Journal*, 131(639):2824–2855.

- Che, Y., Lin, D., and Zhang, Y. (2025). Pains or gains: Trade war, trade deficit, and tariff evasion. *Journal of International Economics*, 155:104090.
- Chor, D. and Li, B. (2024). Illuminating the effects of the us-china tariff war on china's economy. *Journal of International Economics*, 150:103926.
- Christophe, S. E., Ferri, M. G., and Angel, J. J. (2004). Short-selling prior to earnings announcements. *The Journal of Finance*, 59(4):1845–1876.
- Christophe, S. E., Ferri, M. G., and Hsieh, J. (2010). Informed trading before analyst downgrades: Evidence from short sellers. *Journal of Financial Economics*, 95(1):85–106.
- Correa, R., Di Giovanni, J., Goldberg, L. S., and Minoiu, C. (2023). *Trade uncertainty and US bank lending*, volume 31860. National Bureau of Economic Research.
- Crosignani, M., Han, L., and Macchiavelli, M. (2024). Navigating geopolitical risk: Evidence from us mutual funds. *Available at SSRN 5087499*.
- Dechow, P. M., Hutton, A. P., Meulbroek, L., and Sloan, R. G. (2001). Short-sellers, fundamental analysis, and stock returns. *Journal of financial Economics*, 61(1):77–106.
- Di Maggio, M., Kermani, A., and Palmer, C. J. (2020). How quantitative easing works: Evidence on the refinancing channel. *The Review of Economic Studies*, 87(3):1498–1528.
- Diether, K. B., Lee, K.-H., and Werner, I. M. (2009). Short-sale strategies and return predictability. *The Review of Financial Studies*, 22(2):575–607.
- Drechsler, I., Savov, A., and Schnabl, P. (2017). The deposits channel of monetary policy. *The Quarterly Journal of Economics*, 132(4):1819–1876.
- Duffie, D. (2010). Asset price dynamics with slow-moving capital,(american finance association presidential address). *Journal of Finance*, 65:1238–68.

- Efing, M., Goldbach, S., and Nitsch, V. (2023). Freeze! financial sanctions and bank responses. *The Review of Financial Studies*, 36(11):4417–4459.
- Fajgelbaum, P., Goldberg, P., Kennedy, P., Khandelwal, A., and Taglioni, D. (2024). The us-china trade war and global reallocations. *American Economic Review: Insights*, 6(2):295–312.
- Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J., and Khandelwal, A. K. (2020). The return to protectionism. *Quarterly Journal of Economics*, 135(1):1–55.
- Flaaen, A., Hortaçsu, A., and Tintelnot, F. (2020). The production relocation and price effects of us trade policy: the case of washing machines. *American Economic Review*, 110(7):2103–2127.
- Gabaix, X. and Koijen, R. S. (2021). In search of the origins of financial fluctuations: The inelastic markets hypothesis. Technical report, National Bureau of Economic Research.
- Giannetti, M. and Saidi, F. (2019). Shock propagation and banking structure. *The Review of Financial Studies*, 32(7):2499–2540.
- Grossman, G. M., Helpman, E., and Redding, S. J. (2024). When tariffs disrupt global supply chains. *American Economic Review*, 114(4):988–1029.
- Hale, G., Kapan, T., and Minoiu, C. (2020). Shock transmission through cross-border bank lending: Credit and real effects. *The Review of Financial Studies*, 33(10):4839–4882.
- Handley, K., Kamal, F., and Monarch, R. (2024). Supply chain adjustments to tariff shocks: Evidence from firm trade linkages in the 2018-2019 us trade war. *Economics Letters*, 244:112009.
- Handley, K., Kamal, F., and Monarch, R. (2025). Rising import tariffs, falling exports: When modern supply chains meet old-style protectionism. *American Economic Journal: Applied Economics*, 17(1):208–238.

- He, Z. and Krishnamurthy, A. (2013). Intermediary asset pricing. *American Economic Review*, 103(2):732–770.
- Hinz, J. and Monastyrchenko, E. (2022). Bearing the cost of politics: Consumer prices and welfare in russia. *Journal of International Economics*, 137:103581.
- Hou, K. and Moskowitz, T. J. (2005). Market frictions, price delay, and the cross-section of expected returns. *The Review of Financial Studies*, 18(3):981–1020.
- Huang, Y., Lin, C., Liu, S., and Tang, H. (2023). Trade networks and firm value: Evidence from the us-china trade war. *Journal of International Economics*, 145:103811.
- Jiang, L., Lu, Y., Song, H., and Zhang, G. (2023). Responses of exporters to trade protectionism: Inferences from the us-china trade war. *Journal of International Economics*, 140:103687.
- Jiao, Y., Liu, Z., Tian, Z., and Wang, X. (2024). The impacts of the us trade war on chinese exporters. *Review of Economics and Statistics*, 106(6):1576–1587.
- Khwaja, A. I. and Mian, A. (2008). Tracing the impact of bank liquidity shocks: Evidence from an emerging market. *American Economic Review*, 98(4):1413–1442.
- Kruttli, M. S., Monin, P. J., and Watugala, S. W. (2022). The life of the counterparty: Shock propagation in hedge fund-prime broker credit networks. *Journal of Financial Economics*, 146(3):965–988.
- Ray, W., Droste, M., and Gorodnichenko, Y. (2024). Unbundling quantitative easing: Taking a cue from treasury auctions. *Journal of Political Economy*, 132(9):3115–3172.

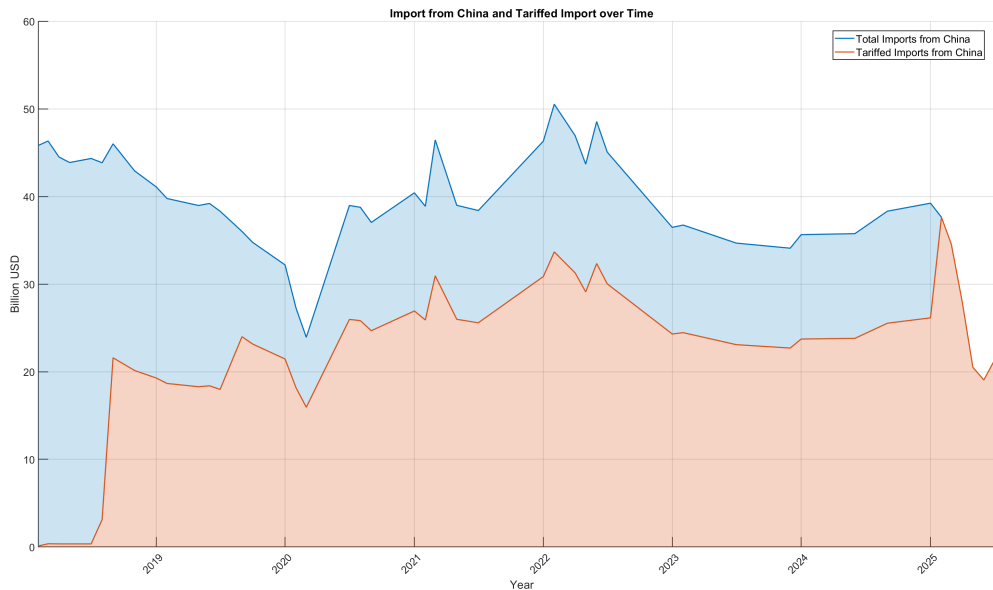


Figure 1: An increasing share of U.S. imports from China is subject to tariffs. Monthly import data are from the U.S. Census Bureau.

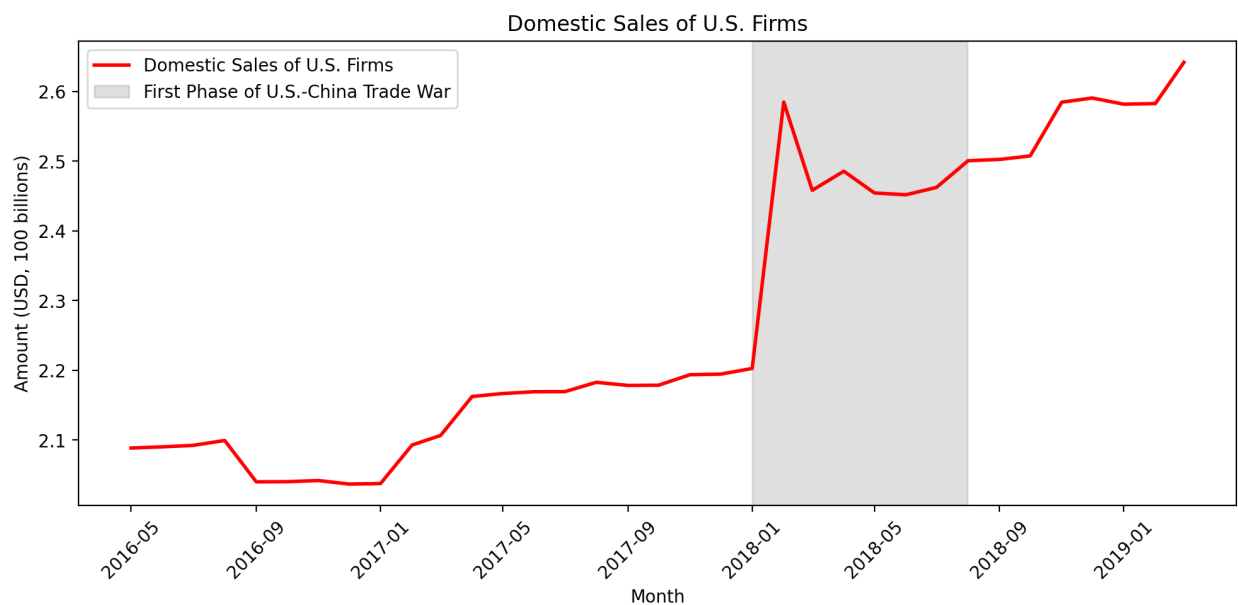


Figure 2: Domestic sales of U.S. firms show a clear upward shift at the beginning of the first U.S.-China trade war, suggesting the firms' reorientation toward the home market. Domestic sales data are from FactSet Revere Geographic Exposure database.

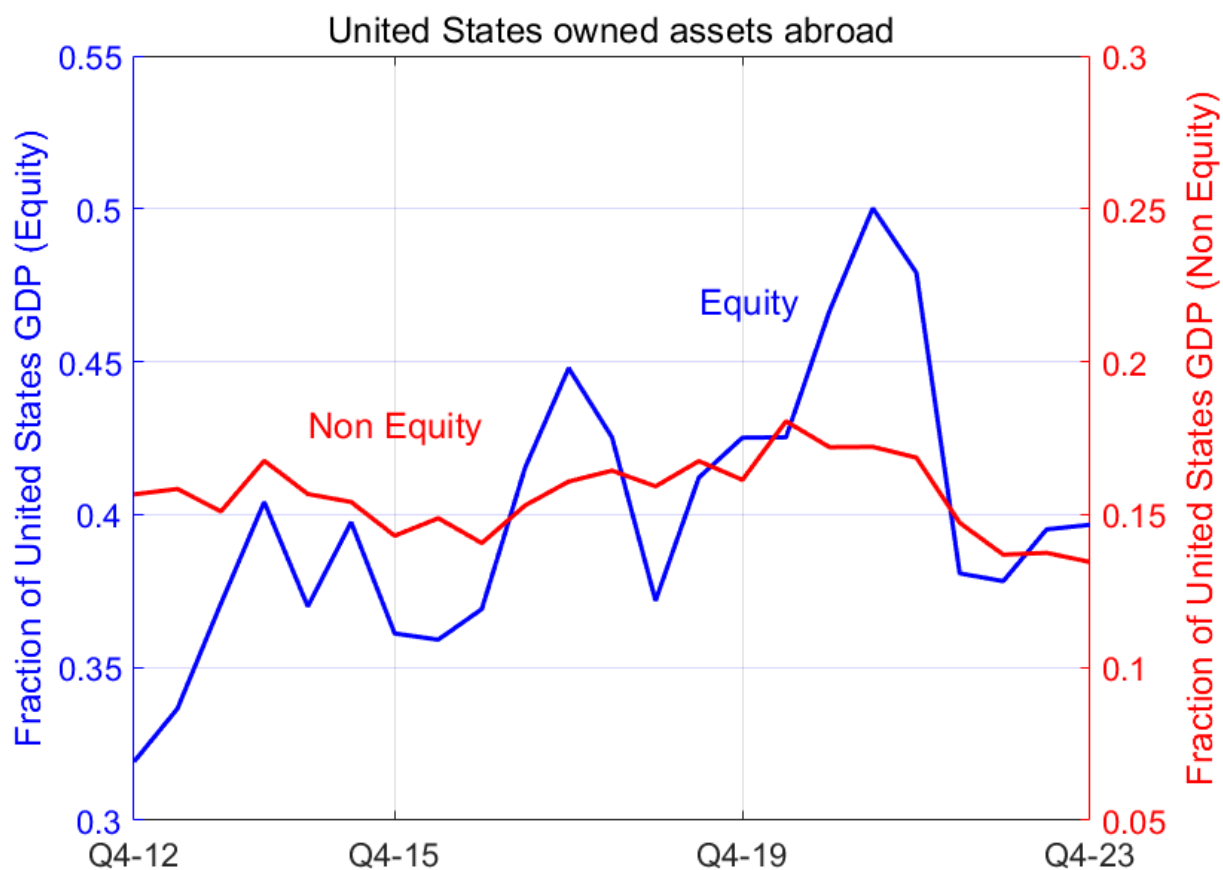


Figure 3: United States owned assets abroad from 2012 to 2023. Data are obtained from the IMF Portfolio Investment Positions by Counterpart Economy dataset (formerly Coordinated Portfolio Investment Survey, or CPIS).

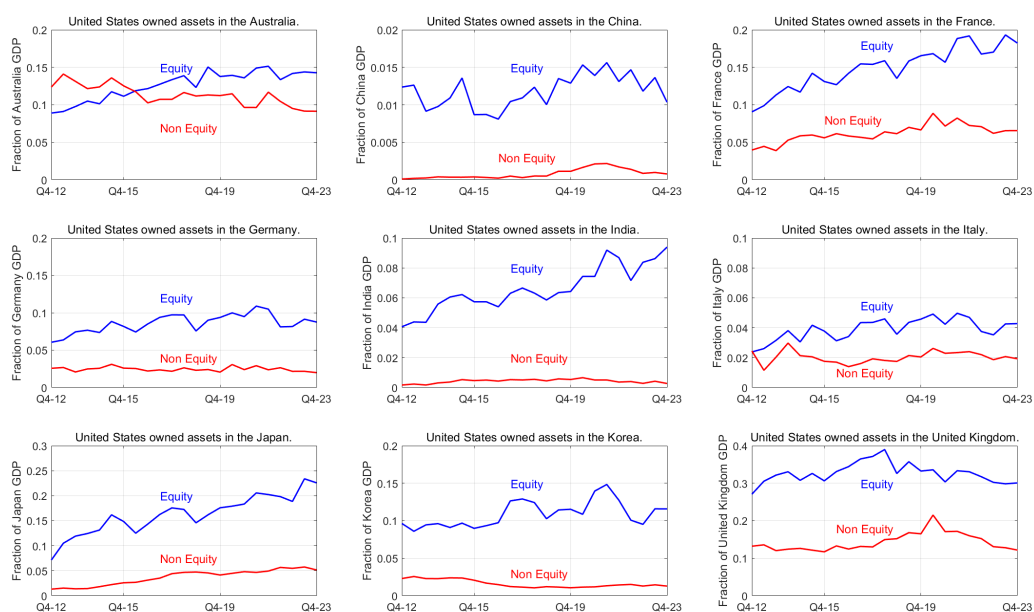


Figure 4: United States owned assets in each country from 2012 to 2023. Data are obtained from the IMF Portfolio Investment Positions by Counterpart Economy dataset (formerly Coordinated Portfolio Investment Survey, or CPIS).

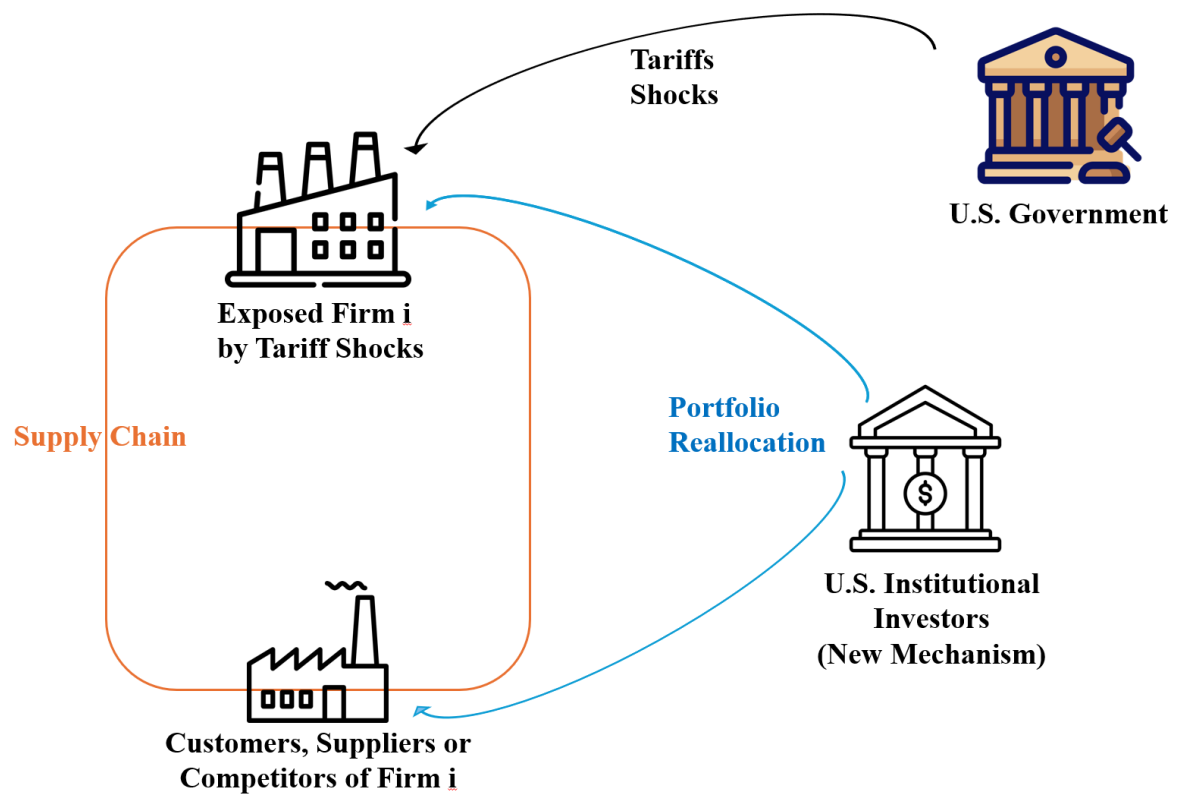


Figure 5: A novel mechanism: how do the U.S. institutional investors amplify the tariff shocks?

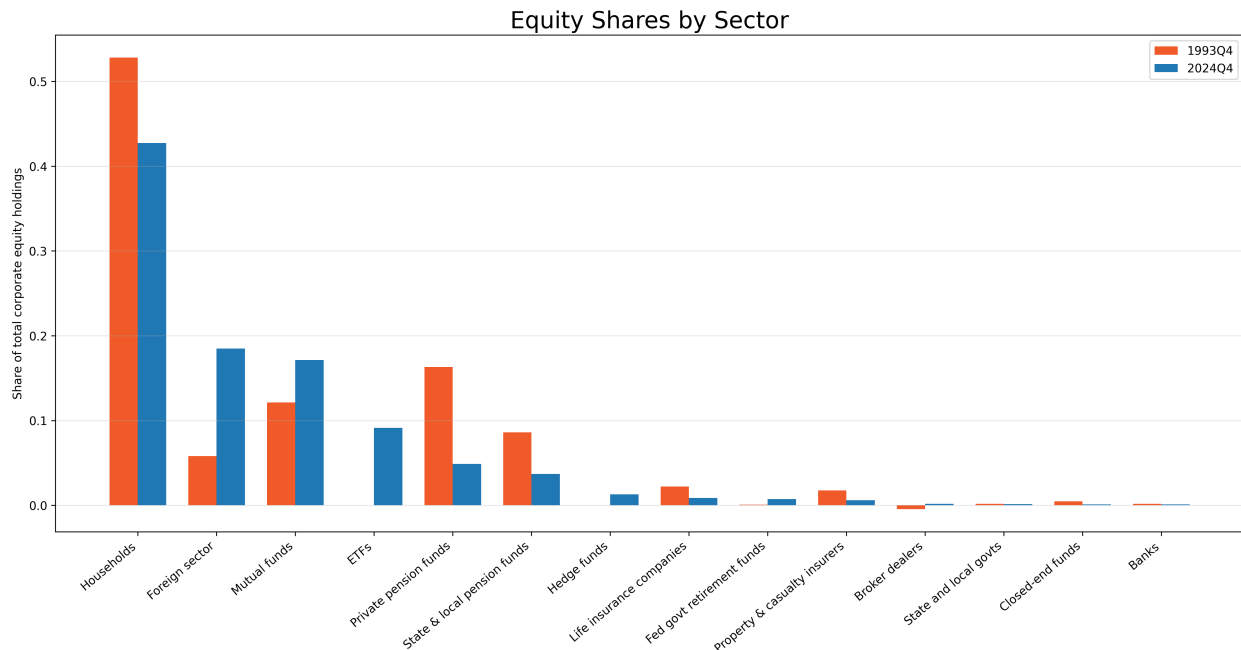


Figure 6: Shares of U.S. corporate equity holdings are constructed using quarterly Flow of Funds data from the Federal Reserve's Financial Accounts of the United States (Z.1 release).

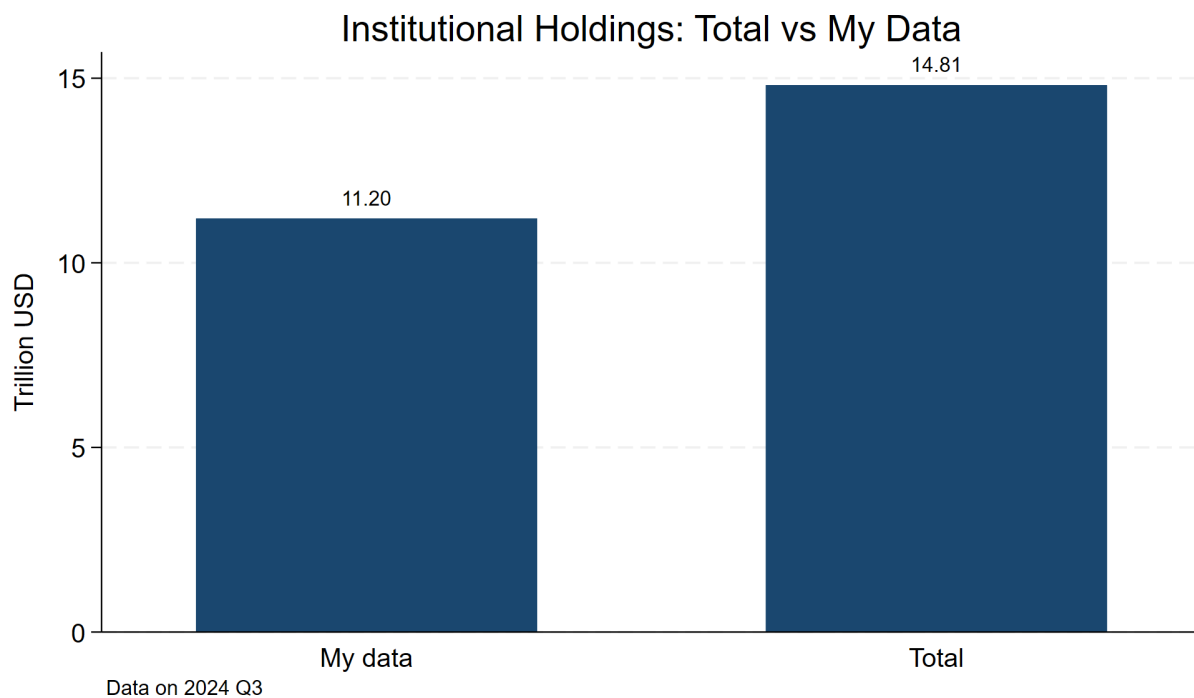


Figure 7: The right-hand amount is composed of equity holdings by hedge funds, mutual funds, insurance companies, and pension funds.

Table 1: Summary Statistics

Variable	Obs	Mean	Std. Dev.	p25	p50	p75
Panel A: U.S. Firms						
PCR	81,908	0.028	0.246	-0.041	0.010	0.095
ROA	79,968	-0.032	0.113	-0.039	0.003	0.017
Leverage Ratio	79,947	0.293	0.283	0.066	0.254	0.423
Cash Ratio	79,966	0.261	0.288	0.040	0.132	0.409
$\Delta \log$ (Sales)	70,361	0.048	0.271	-0.056	0.023	0.110
Net Margin	74,150	0.001	0.228	-0.044	0.034	0.099
$\Delta \log$ (R&D)	36,095	0.010	0.226	-0.062	0.026	0.117
Tobin's Q	75,617	1.939	1.467	0.936	1.416	2.443
$\Delta \log$ (Capex)	37,861	-0.008	0.317	-0.081	0.019	0.117
Panel B: Foreign Firms						
PCR	175,654	0.021	0.200	0.000	0.000	0.037
ROA	167,728	0.011	0.059	0.000	0.001	0.010
Leverage Ratio	167,549	0.273	0.833	0.006	0.025	0.134
Cash Ratio	166,740	0.251	0.701	0.008	0.026	0.160
$\Delta \log$ (Sales)	163,726	0.016	0.295	-0.085	0.023	0.129
Net Margin	169,684	0.006	0.050	0.000	0.001	0.010
$\Delta \log$ (R&D)	90,769	0.019	0.523	-0.138	0.020	0.185
Tobin's Q	160,753	1.766	1.851	0.739	1.122	1.987
$\Delta \log$ (Capex)	136,339	0.019	0.968	-0.434	0.020	0.462

Notes. This table 1 reports summary statistics for the main firm-level variables. PCR measures the quarterly change in institutional ownership. $\Delta \log$ Sales and $\Delta \log$ R&D capture quarterly growth in sales and R&D expenditures, respectively. Net Margin is defined as net income divided by sales, and ROA (%) as net income divided by total assets. Tobin's Q is measured as the ratio of the market value of assets—defined as the market value of equity plus preferred stock and debt—to the book value of total assets. Leverage Ratio (%) is total debt divided by total assets, and Cash Ratio (%) is cash holdings divided by total assets; all fundamentals are obtained from the FactSet Fundamentals (North America and International coverage).

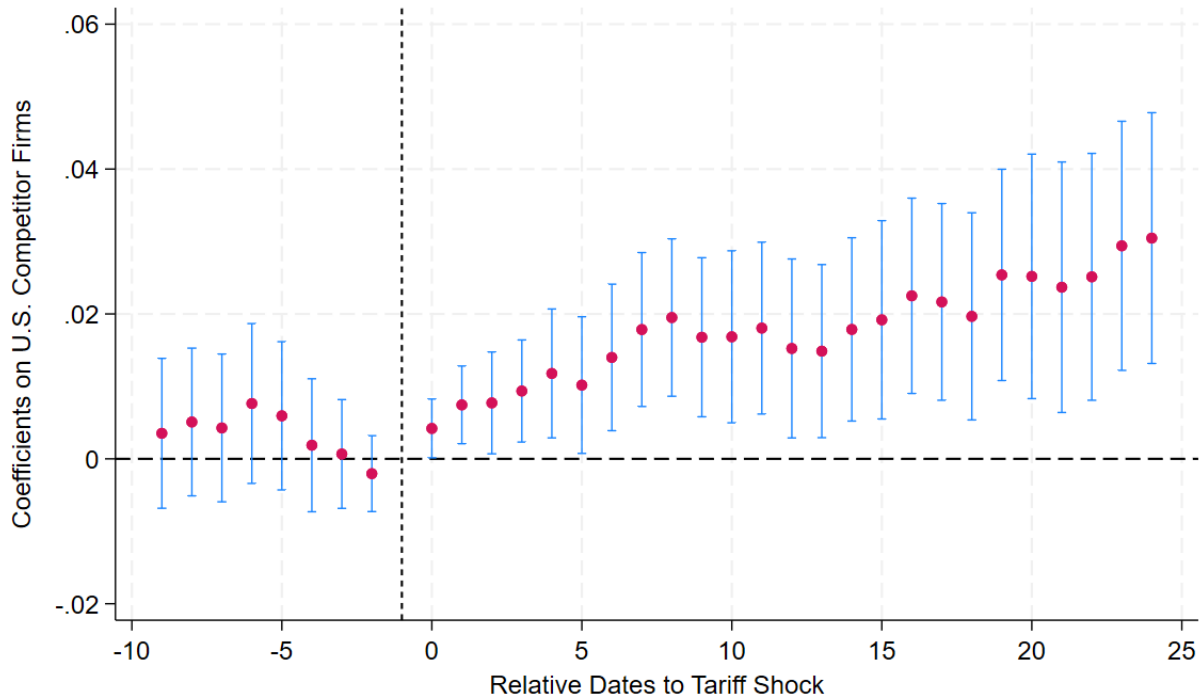


Figure 8: Cumulative abnormal returns (CARs) of U.S. competitor firms increase following U.S. tariff shocks. Abnormal returns are measured as residuals from the Fama–French three-factor model and aggregated within each event window. The sample is restricted to firms that share the same three-digit SIC industry classification as the competitor firms. Standard errors are clustered at the firm level. Error bars report 90% confidence intervals.

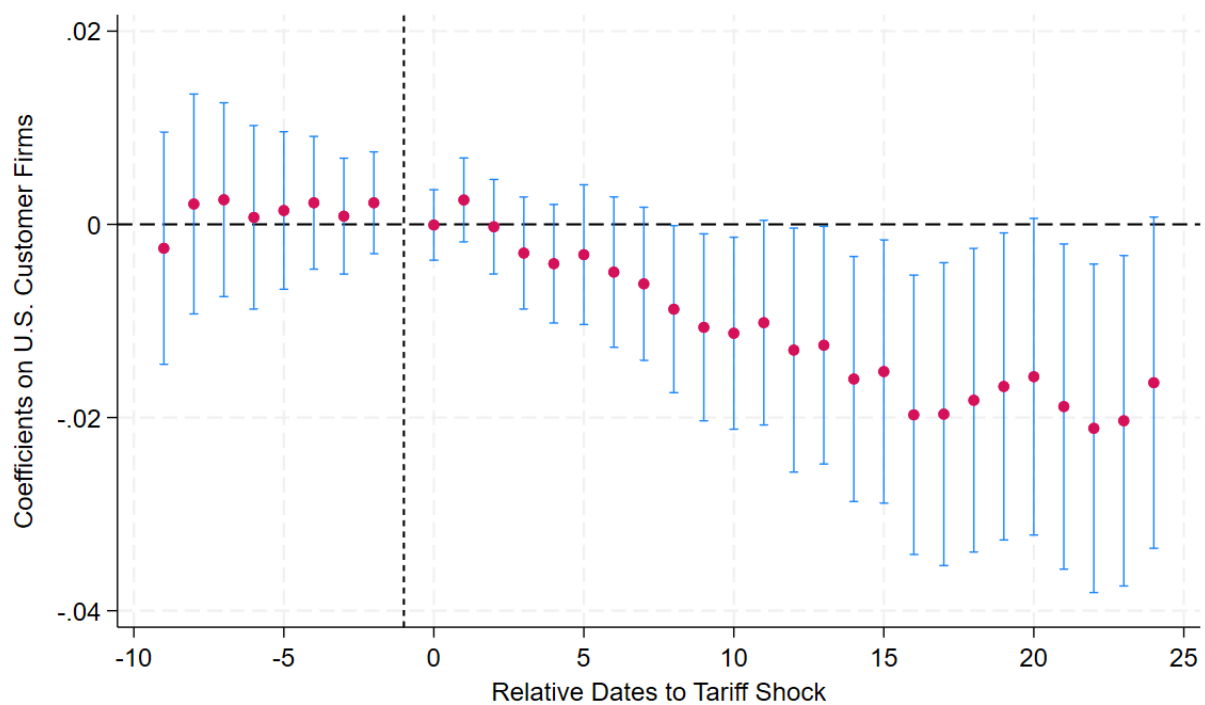


Figure 9: U.S. customer firms suffer from cumulative abnormal return (CAR) decline. Abnormal returns are measured as residuals from the Fama–French three-factor model and aggregated within each event window. The sample is restricted to firms that share the same three-digit SIC industry classification as the shocked customer firms. Standard errors are clustered at the firm level. Error bars report 90% confidence intervals.

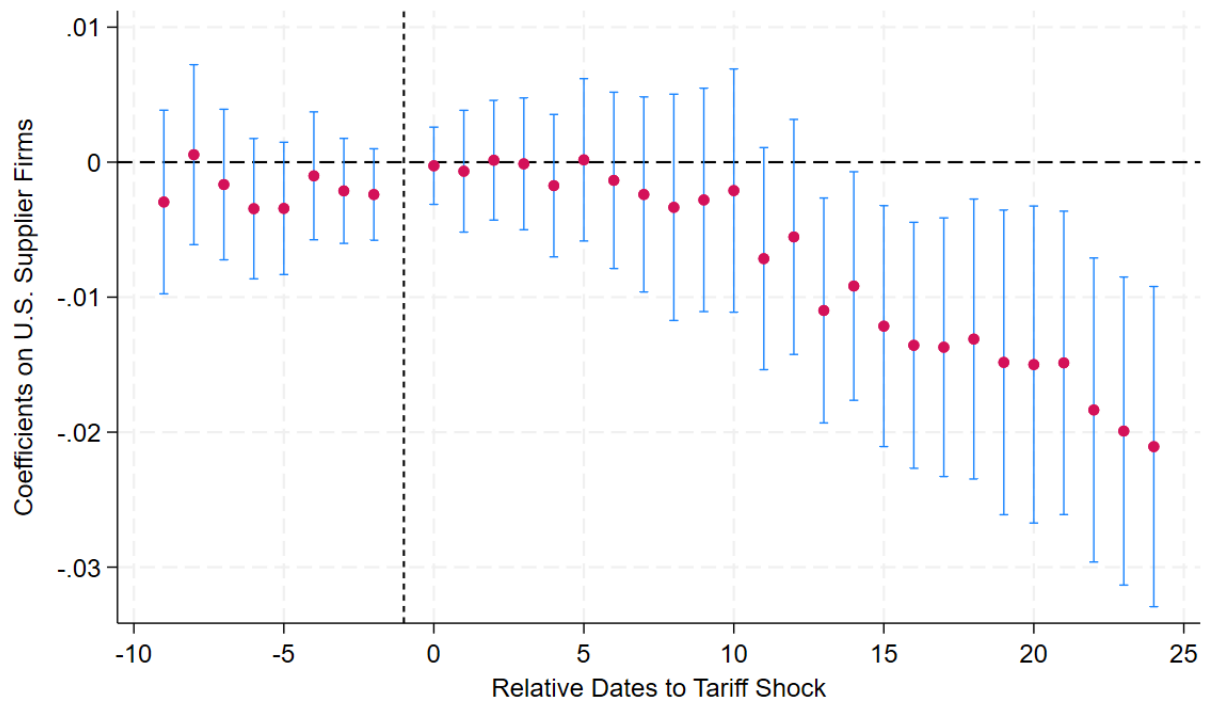


Figure 10: U.S. supplier firms also suffer from cumulative abnormal return (CAR) decline. Abnormal returns are measured as residuals from the Fama–French three-factor model and aggregated within each event window. The sample is restricted to firms that share the same three-digit SIC industry classification as the shocked supplier firms. Standard errors are clustered at the firm level. Error bars report 90% confidence intervals.

Table 2: Quarterly Results for Tariff Shock Measures on PCR

	U.S. Firms				Foreign Firms		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dep. variable: PCR							
Customer $_{i,t}^d$	-0.02607*						
	(0.01411)						
Supplier $_{i,t}^d$		-0.03028**					
		(0.01338)					
Competitor $_{i,t}^d$			0.0210*	0.00789			
			(0.0123)	(0.0119)			
Exposure $_{i,t}^j$					-0.03496**		
					(0.01710)		
Customer $_{i,t}^j$						-0.02725**	
						(0.01276)	
Competitor $_{i,t}^j$							0.0124*
							(0.00691)
R-squared	0.1055	0.1055	0.0986	0.1270	0.1280	0.1280	0.1899
Observations	76,209	76,209	70,994	70,994	145,216	145,216	134,806
Firm Controls:	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Quarter FE:	Yes	Yes	No	Yes	Yes	Yes	Yes

Notes. Table 2 reports quarterly regression results on how U.S. institutional investors adjust their holdings in firm i in response to U.S. tariff shocks. The dependent variable is PCR , defined as the quarterly change ratio in U.S. institutional investors' positions in firm i located in the United States d or foreign country j at time t . *Customer* equals one if a firm is connected as a customer to at least one tariff-exposed firm in quarter t . *Supplier* equals one if a U.S. firm is connected as a supplier to at least one tariff-exposed firm in quarter t . *Exposure* is an indicator equal to one if a foreign firm is directly targeted by a U.S. tariff shock in quarter t . *Competitor* is an indicator equal to one if a firm has a competitor relationship with at least one tariff-exposed firm in quarter t , based on FactSet Revere relationships data. Columns (1)–(4) report results for U.S. firms, and Columns (5)–(7) report results for foreign firms. All regressions control for firm-level characteristics and include firm fixed effects. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 3: Quarterly Results for U.S. Tariff Shocks on U.S. Customer Firms

	(1) Sales Growth	(2) Profitability	(3) R&D Growth	(4) CapEx Growth
$PCR_{i,t}^d$	0.0166*** (0.00319)	-0.00670 (0.141)	0.00325 (0.00304)	0.00492 (0.00323)
$PCR_{i,t-1}^d$	0.0106*** (0.00321)	-0.166 (0.134)	0.0151*** (0.00261)	0.0163*** (0.00301)
$PCR_{i,t-2}^d$	0.0132*** (0.00289)	0.162 (0.112)	0.00850*** (0.00257)	0.0116*** (0.00275)
$Customer_{i,t}^d \times PCR_{i,t}^d$	0.0899** (0.0441)	-0.0461 (0.296)	0.0976* (0.0576)	0.0754 (0.0635)
$Customer_{i,t-1}^d \times PCR_{i,t-1}^d$	-0.0355 (0.0482)	0.282 (0.292)	-0.0578 (0.0595)	-0.0914 (0.0564)
$Customer_{i,t-2}^d \times PCR_{i,t-2}^d$	0.0240 (0.0437)	0.165 (0.274)	0.108** (0.0455)	0.146** (0.0602)
$Customer_{i,t}^d$	0.0154 (0.00959)	0.00698 (0.0590)	0.00310 (0.00907)	0.00199 (0.00985)
$Customer_{i,t-1}^d$	-0.000324 (0.00779)	-0.108** (0.0543)	-0.00787 (0.0114)	-0.0120 (0.0115)
$Customer_{i,t-2}^d$	-0.00420 (0.00581)	-0.0549 (0.0474)	0.000499 (0.0113)	0.00220 (0.0120)
R-squared	0.119	0.621	0.114	0.477
Observations	60214	60294	30243	31804
Firm Controls:	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes
Year-Quarter FE:	Yes	Yes	Yes	Yes

Notes. Table 3 reports quarterly regression results for U.S. firms indirectly affected by tariff shocks. The dependent variables are growth ratio of sales (column 1), net profit margin (column 2), growth ratio of R&D (column 3), and growth rate of Capital Expenditure (column 4). *Customer* is an indicator variable equal to one if a U.S. firm is connected through the supply chain to at least one exposed firm and serves as its customer in quarter t . *PCR*, measures the quarterly change ratio of positions in firm i located in the U.S. d at time t held by U.S. institutional Investors. All regressions control for firm-level characteristics and include firm and year-quarter fixed effects. Standard errors clustered at the firm level are presented. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 4: Quarterly Results for U.S. Tariff Shocks on U.S. Suppliers

	(1) Sales Growth	(2) Profitability	(3) R&D Growth	(4) Tobin's Q
$PCR_{i,t}^d$	0.0212*** (0.00526)	0.00514** (0.00219)	0.00576 (0.00375)	0.0233*** (0.00616)
$PCR_{i,t-1}^d$	0.0127** (0.00546)	0.00514** (0.00257)	0.0181*** (0.00322)	0.111*** (0.00662)
$PCR_{i,t-2}^d$	0.0230*** (0.00558)	0.00394** (0.00192)	0.0108*** (0.00302)	0.0679*** (0.00579)
$Supplier_{i,t}^d \times PCR_{i,t}^d$	0.0404 (0.0659)	0.0572** (0.0283)	-0.0532 (0.0446)	0.0882 (0.111)
$Supplier_{i,t-1}^d \times PCR_{i,t-1}^d$	-0.0548 (0.0592)	0.0437 (0.0400)	0.00479 (0.0706)	0.172* (0.0966)
$Supplier_{i,t-2}^d \times PCR_{i,t-2}^d$	-0.0172 (0.0965)	0.0316 (0.0345)	-0.0381 (0.0492)	0.308** (0.146)
$Supplier_{i,t}^d$	0.0142 (0.0123)	-0.00204 (0.00539)	0.0114 (0.00895)	-0.0652*** (0.0225)
$Supplier_{i,t-1}^d$	-0.00431 (0.0113)	0.00354 (0.00636)	-0.0151 (0.0112)	-0.0292* (0.0177)
$Supplier_{i,t-2}^d$	-0.00351 (0.00888)	-0.00352 (0.00702)	-0.00956 (0.00965)	-0.0146 (0.0220)
R-squared	0.132	0.537	0.0886	0.772
Observations	60214	54317	30243	58650
Firm Controls:	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes
Year-Quarter FE:	Yes	Yes	Yes	Yes

Notes. Table 4 reports quarterly regression results for U.S. supplier firms. The dependent variables are growth ratio of sales (column 1), net profit margin (column 2), growth ratio of R&D (column 3), and Tobin's Q (column 4). *Supplier* is an indicator variable equal to one if a U.S. firm is connected through the supply chain to at least one exposed firm and serves as its supplier in quarter t . *PCR*, measures the quarterly change ratio of positions in firm i located in country j at time t held by U.S. institutional Investors. All regressions control for firm-level characteristics and include firm and year-quarter fixed effects. Standard errors clustered at the firm level are presented. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Quarterly Results for U.S. Tariff Shocks on U.S. Competitors

	(1) Sales Growth	(2) Profitability	(3) R&D Growth	(4) CapEx Growth
$PCR_{i,t}^d$	0.0155*** (0.00284)	-0.0016 (0.0749)	0.00565 (0.00373)	0.00412 (0.00277)
$PCR_{i,t-1}^d$	0.00992*** (0.00287)	-0.098 (0.0775)	0.0182*** (0.00321)	0.0144*** (0.00260)
$PCR_{i,t-2}^d$	0.0120*** (0.00255)	0.0265 (0.0711)	0.0108*** (0.00301)	0.0102*** (0.00238)
$Competitor_{i,t}^d \times PCR_{i,t}^d$	0.0153 (0.0438)	-0.0086 (0.113)	0.00670 (0.0967)	-0.0323 (0.0907)
$Competitor_{i,t-1}^d \times PCR_{i,t-1}^d$	0.0484 (0.0449)	0.278* (0.153)	0.191 (0.281)	0.0717 (0.213)
$Competitor_{i,t-2}^d \times PCR_{i,t-2}^d$	-0.0190 (0.0441)	-0.076 (0.114)	0.245 (0.228)	0.226** (0.115)
$Competitor_{it}^d$	0.0231* (0.0130)	0.0543* (0.0327)	0.0194 (0.0259)	0.0299 (0.0288)
$Competitor_{i,t-1}^d$	-0.0238 (0.0146)	-0.0279 (0.0301)	-0.0260 (0.0338)	-0.0137 (0.0316)
$Competitor_{i,t-2}^d$	0.00596 (0.0116)	0.0211 (0.0267)	0.0268 (0.0381)	0.0217 (0.0311)
R-squared	0.119	0.707	0.0886	0.438
Observations	60214	60294	30243	31804
Firm Controls:	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes
Year-Quarter FE:	Yes	Yes	Yes	Yes

Notes. Table 5 reports quarterly regression results for U.S. competitor firms. The dependent variables are growth ratio of sales (column 1), net profit margin (column 2), growth ratio of R&D (column 3), and growth ratio of Capital Expenditure (column 4). *Competitor* is an indicator variable equal to one if a U.S. firm has a competitor link with at least one tariff-exposed firm in time t based on FactSet Revere relationships data. *PCR*, measures the quarterly change ratio of positions in firm i located in country j at time t held by U.S. institutional Investors. All regressions control for firm-level characteristics and include firm and year-quarter fixed effects. Standard errors clustered at the firm level are presented. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 6: Quarterly Results for U.S. Tariff Shocks on Foreign Exposed Firms

	(1) Sales Growth	(2) Profitability	(3) R&D Growth	(4) Tobin's Q
$PCR_{i,t}^j$	0.00371* (0.00190)	0.00145*** (0.000281)	0.00878* (0.00507)	0.0172*** (0.00605)
$PCR_{i,t-1}^j$	0.00239 (0.00182)	0.00130*** (0.000299)	-0.00140 (0.00454)	0.0570*** (0.00671)
$PCR_{i,t-2}^j$	0.0104*** (0.00153)	0.000944*** (0.000224)	0.0257*** (0.00491)	0.0545*** (0.00570)
$Exposure_{i,t}^j \times PCR_{i,t}^j$	0.0747 (0.0994)	0.00228 (0.00234)	0.476*** (0.0872)	0.0273 (0.0493)
$Exposure_{i,t-1}^j \times PCR_{i,t-1}^j$	-0.0365 (0.0375)	-0.0000481 (0.00381)	-0.331 (0.205)	-0.0592 (0.0496)
$Exposure_{i,t-2}^j \times PCR_{i,t-2}^j$	0.0293 (0.0485)	0.00127 (0.00227)	0.0333 (0.106)	-0.0529 (0.0681)
$Exposure_{i,t}^j$	0.00140 (0.0138)	0.00104 (0.00135)	0.00368 (0.0428)	-0.200*** (0.0240)
$Exposure_{i,t-1}^j$	-0.0185 (0.0192)	-0.00314 (0.00284)	-0.0113 (0.0450)	-0.149*** (0.0235)
$Exposure_{i,t-2}^j$	0.0157 (0.0176)	0.000627 (0.00150)	-0.0339 (0.0415)	-0.0931*** (0.0268)
R-squared	0.0980	0.489	0.112	0.849
Observations	120699	118445	72200	113907
Firm Controls:	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes
Year-Quarter FE:	Yes	Yes	Yes	Yes

Notes. Table 6 reports quarterly regression results for foreign firms directly affected by tariff shocks. The dependent variables are growth ratio of sales (column 1), net profit margin (column 2), growth ratio of R&D (column 3), and Tobin's Q (column 4). *Exposure* is an indicator variable equal to one if a firm is directly targeted by U.S. tariff shocks in t . *PCR*, measures the quarterly change ratio of positions on firm i located in country j at t held by U.S. institutional Investors. All regressions control for firm-level characteristics and include firm and year-quarter fixed effects. Standard errors clustered at the firm level are presented. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 7: Quarterly Results for U.S. Tariff Shocks on Foreign Spillover Firms

	(1) Sales Growth	(2) Profitability	(3) R&D Growth	(4) Tobin's Q
$PCR_{i,t}^j$	0.00384** (0.00191)	0.00143*** (0.000282)	0.00906* (0.00510)	0.0170*** (0.00607)
$PCR_{i,t-1}^j$	0.00242 (0.00183)	0.00126*** (0.000298)	-0.00184 (0.00456)	0.0564*** (0.00672)
$PCR_{i,t-2}^j$	0.0104*** (0.00153)	0.000941*** (0.000225)	0.0254*** (0.00491)	0.0545*** (0.00571)
$Spillover_{i,t}^j \times PCR_{i,t}^j$	-0.00481 (0.0183)	0.00359 (0.00341)	0.0391 (0.0503)	0.0417 (0.0450)
$Spillover_{i,t-1}^j \times PCR_{i,t-1}^j$	-0.0177 (0.0169)	0.0101*** (0.00377)	-0.0196 (0.0614)	0.153*** (0.0479)
$Spillover_{i,t-2}^j \times PCR_{i,t-2}^j$	0.00282 (0.0227)	0.00257 (0.00215)	0.0878* (0.0519)	0.00936 (0.0387)
$Spillover_{i,t}^j$	0.00378 (0.00801)	-0.00294* (0.00161)	0.00699 (0.0184)	-0.0575*** (0.0138)
$Spillover_{i,t-1}^j$	-0.00543 (0.00804)	-0.000453 (0.00130)	0.00376 (0.0190)	-0.00990 (0.0133)
$Spillover_{i,t-2}^j$	-0.000638 (0.00692)	0.000645 (0.000977)	-0.0428** (0.0184)	-0.0327** (0.0165)
R-squared	0.0980	0.489	0.112	0.849
Observations	120699	118445	72200	113907
Firm Controls:	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes
Year-Quarter FE:	Yes	Yes	Yes	Yes

Notes. Table 7 reports quarterly regression results for foreign firms indirectly affected by tariff shocks, or spillover firms. The dependent variables are growth ratio of sales (column 1), net profit margin (column 2), growth ratio of R&D (column 3), and Tobin's Q (column 4). *Customer* is an indicator variable equal to one if a foreign firm is connected through the supply chain to at least one exposed firm and serves as its customer in quarter t . *PCR*, measures the quarterly change ratio of positions on firm i located in country j at t held by U.S. institutional Investors. All regressions control for firm-level characteristics and include firm and year-quarter fixed effects. Standard errors clustered at the firm level are presented. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 8: Quarterly Results for U.S. Tariff Shocks on Foreign Competitors

	(1) Sales Growth	(2) Profitability	(3) R&D Growth	(4) Tobin's Q
$PCR_{i,t}^j$	0.00382** (0.00190)	0.00144*** (0.000280)	0.00902* (0.00508)	0.0170*** (0.00605)
$PCR_{i,t-1}^j$	0.00232 (0.00182)	0.00129*** (0.000298)	-0.00178 (0.00454)	0.0568*** (0.00670)
$PCR_{i,t-2}^j$	0.0105*** (0.00153)	0.000941*** (0.000224)	0.0258*** (0.00491)	0.0544*** (0.00570)
$Competitor_{i,t}^j \times PCR_{i,t}^j$	-0.0657 (0.118)	0.0183 (0.0159)	0.0505 (0.144)	0.652** (0.301)
$Competitor_{i,t-1}^j \times PCR_{i,t-1}^j$	0.0954 (0.0834)	0.0111 (0.0133)	-0.165 (0.198)	0.490 (0.326)
$Competitor_{i,t-2}^j \times PCR_{i,t-2}^j$	-0.0355 (0.0345)	0.00882 (0.00883)	-0.164 (0.204)	0.199 (0.217)
$Competitor_{i,t}^j$	0.00902 (0.0131)	0.00241 (0.00226)	0.00571 (0.0265)	-0.0231 (0.0249)
$Competitor_{i,t-1}^j$	0.00233 (0.0143)	0.00120 (0.00211)	-0.0273 (0.0339)	0.0394 (0.0336)
$Competitor_{i,t-2}^j$	-0.00937 (0.0116)	-0.000510 (0.00257)	-0.0296 (0.0304)	0.0473 (0.0323)
R-squared	0.0980	0.489	0.112	0.849
Observations	120699	118445	72200	113907
Firm Controls:	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes
Year-Quarter FE:	Yes	Yes	Yes	Yes

Notes. Table 8 reports quarterly regression results for foreign competitor firms. The dependent variables are growth ratio of sales (column 1), net profit margin (column 2), growth ratio of R&D (column 3), and Tobin's Q (column 4). *Competitor* is an indicator variable equal to one if a foreign firm has a competitor link with at least one tariff-exposed firm in time t based on FactSet Revere relationships data. *PCR*, measures the quarterly change ratio of positions in firm i located in country j at t held by U.S. institutional Investors. All regressions control for firm-level characteristics and include firm and year-quarter fixed effects. Standard errors clustered at the firm level are presented. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 9: Annual Results for U.S. Tariff Shocks on Foreign Sales Growth

	Foreign Sales Growth (%)			
	(1)	(2)	(3)	(4)
Supplier $_{i,t}^d$	-4.735 (3.338)	-2.563 (2.592)		
Supplier $_{i,t-1}^d$	-6.814* (3.535)	-4.675* (2.819)		
Customer $_{i,t}^d$			-0.459 (2.282)	-1.664 (2.259)
Customer $_{i,t-1}^d$			-4.483** (1.964)	-3.289* (1.781)
Firm FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
Sector $\times$ Year FE	Yes	No	Yes	No
R-squared	0.144	0.347	0.131	0.287
Observations	11,587	11,808	11,578	11,811

Notes. Table 9 reports annual regression results for the effect of U.S. tariff shocks on firms' foreign sales growth. The dependent variable is foreign sales growth, measured using customer segment information from WRDS Compustat Historical Segments. *Supplier* is an indicator variable equal to one if a U.S. firm is connected through the supply chain to at least one exposed firm and serves as its supplier in year t . *Customer* is an indicator variable equal to one if a U.S. firm is connected through the supply chain to at least one exposed firm and serves as its customer in year t . Columns (1) and (3) include sector-by-year fixed effects, while columns (2) and (4) include firm and year fixed effects. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 10: Effect of Domestic Suppliers on Firm Performances During The First U.S. China Trade War

	(1) Sales Growth	(2) Profitability	(3) Tobin's Q	(4) Sales Growth	(5) Profitability	(6) Tobin's Q
TW_{t-1}	-0.0229*** (0.00737)	-0.00706** (0.00307)	-0.0445*** (0.0101)	-0.0301*** (0.00823)	-0.00853** (0.00348)	-0.0465*** (0.0118)
$TW_{t-1} \cdot \text{Simple Avg}_{S,i,t-1}^{d,supplier}$	0.0116 (0.0145)	0.00573 (0.00613)	0.0335* (0.0187)			
$TW_{t-1} \cdot \text{Weighted Avg}_{S,i,t-1}^{d,supplier}$				0.0273* (0.0155)	0.00851 (0.00697)	0.0338 (0.0224)
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.123	0.548	0.783	0.122	0.549	0.784
Observations	31622	29008	29065	31544	28949	28985

Notes. Table 10 reports quarterly regression results for the effect of domestic suppliers on U.S. firms' performance during the first U.S.-China trade war. The dependent variables are sales growth, profitability, and Tobin's Q. TW_{t-1} is an indicator variable equal to one for quarters during the U.S.-China trade war period, from 2018Q1 to 2019Q4, and zero otherwise. Simple $\text{Avg}_{S,i,t-1}^{d,supplier}$ measures the share of firm i 's suppliers headquartered in the United States in quarter $t - 1$, while Weighted $\text{Avg}_{S,i,t-1}^{d,supplier}$ measures the share of firm i 's suppliers' sales conducted in the United States in quarter $t - 1$. The interaction terms capture whether firms with more domestically oriented supplier networks experienced different performance changes during the trade war period. All regressions control for firm-level characteristics and include firm fixed effects. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 11: Effect of Domestic Customers on Firm Performances During The First U.S. China Trade War

	(1) Sales Growth	(2) Profitability	(3) Tobin's Q	(4) Sales Growth	(5) Profitability	(6) Tobin's Q
TW_{t-1}	-0.0211*** (0.00343)	-0.00682*** (0.00239)	-0.0531*** (0.0117)	-0.0159*** (0.00439)	-0.00711** (0.00300)	-0.0794*** (0.0146)
$TW_{t-1} \cdot \text{Simple Avg}_{i,t-1}^{d,customer}$	0.0164** (0.00694)	0.00637 (0.00486)	0.122*** (0.0249)			
$TW_{t-1} \cdot \text{Weighted Avg}_{i,t-1}^{d,customer}$				0.00143 (0.0102)	0.00630 (0.00696)	0.176*** (0.0325)
Firm Controls:	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.139	0.581	0.801	0.139	0.582	0.801
Observations	30226	28232	30191	30188	28208	30142

Notes. Table 11 reports quarterly regression results for the effect of domestic customers on U.S. firms' performance during the first U.S.-China trade war. The dependent variables are sales growth, profitability, and Tobin's Q. TW_{t-1} is an indicator variable equal to one for quarters during the U.S.-China trade war period, from 2018Q1 to 2019Q4, and zero otherwise. Simple $\text{Avg}_{i,t-1}^{d,customer}$ measures the share of firm i 's customers headquartered in the United States in quarter $t - 1$, while $\text{Weighted Avg}_{i,t-1}^{d,customer}$ measures the share of firm i 's customers' sales conducted in the United States in quarter $t - 1$. The interaction terms capture whether firms with more domestically oriented customer networks experienced different performance changes during the trade war period. All regressions control for firm-level characteristics and include firm fixed effects. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 12: Quarterly Results for U.S. Tariff Shocks on Supply Chain Re-organization

	Supplier Domesticity		Customer Domesticity	
	(1) Simple Avg.	(2) Weighted Avg.	(3) Simple Avg.	(4) Weighted Avg.
$\text{Supplier}_{i,t}^d$	0.0333*** (0.0096)	0.0136** (0.0069)		
$\text{Supplier}_{i,t-1}^d$	0.0316*** (0.0087)	0.0267*** (0.0070)		
$\text{Customer}_{i,t}^d$			0.0197** (0.0082)	0.0007 (0.0066)
$\text{Customer}_{i,t-1}^d$			0.0157** (0.0079)	-0.0012 (0.0070)
Firm Controls:	Yes	Yes	Yes	Yes
Firm FE:	Yes	Yes	Yes	Yes
R-squared	0.6665	0.6664	0.6850	0.6576
Observations	33,227	33,154	33,266	33,224

Notes. Table 12 reports quarterly regression results for U.S. firms' supply chain re-organization following tariff shocks. The dependent variables are measures of supply chain partner domesticity. *Supplier Domesticity* measures the domestic orientation of a firm's suppliers: the simple-average measure is the share of suppliers headquartered in the United States, while the weighted-average measure is the share of suppliers' sales conducted in the United States. *Customer Domesticity* is defined analogously for a firm's customers: the simple-average measure is the share of customers headquartered in the United States, while the weighted-average measure is the share of customers' sales conducted in the United States. $\text{Supplier}_{i,t}^d$ is an indicator variable equal to one if firm i is connected through the supply chain to at least one exposed firm and serves as its supplier in quarter t . $\text{Customer}_{i,t}^d$ is an indicator variable equal to one if firm i is connected through the supply chain to at least one exposed firm and serves as its customer in quarter t . All regressions control for firm-level characteristics and include firm fixed effects. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 13: Quarterly Results for U.S.-China Trade War on Supply Chain Re-organization

	Supplier Domesticity		Customer Domesticity	
	(1)	(2)	(3)	(4)
	Simple Avg.	Weighted Avg.	Simple Avg.	Weighted Avg.
TW_{t-1}	0.0045 (0.0049)	0.0205*** (0.0040)	0.0172*** (0.0054)	0.0113*** (0.0039)
Firm Controls:	Yes	Yes	Yes	Yes
Sector×Year-Quarter FE:	Yes	Yes	Yes	Yes
R-squared	0.6662	0.6671	0.6853	0.6580
Observations	33,227	33,154	33,266	33,224

Notes. Table 13 reports quarterly regression results for U.S. firms' supply chain re-organization during the U.S.-China trade war. The dependent variables are measures of supply chain partner domesticity. *Supplier Domesticity* measures the domestic orientation of a firm's suppliers: the simple-average measure is the share of suppliers headquartered in the United States, while the weighted-average measure is the share of suppliers' sales conducted in the United States. *Customer Domesticity* is defined analogously for a firm's customers: the simple-average measure is the share of customers headquartered in the United States, while the weighted-average measure is the share of customers' sales conducted in the United States. TW_{t-1} is an indicator variable equal to one for quarters during the U.S.-China trade war period, from 2018Q1 to 2019Q4, and zero otherwise. All regressions control for firm-level characteristics and include sector-by-year-quarter fixed effects. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.